

Statistical modeling

Module 01

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Bachelor, GSEM

Autumn 2018

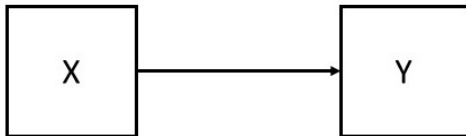
Agenda

- ▶ What is a statistical model?
- ▶ Reminders of univariate statistics
- ▶ Correlation
- ▶ Other dependence concepts

Introduction: What is a model?

A model is a **simplified and imperfect** representation of the reality.

In this course, a link description between a response Y and some characteristics X .



Introduction: What is a model?

Examples of y explained by x 's:

- ▶ Supply chain: demand explained by time (season), special events, other variables.
- ▶ Banking: loan refunding explained by personal customer characteristics.
- ▶ Marketing: product buying chance explained by previous buying habits, personal characteristics...
- ▶ HR: burn out risk explained by personal characteristics.
- ▶ Psychology: behavior explained by personal preferences.
- ▶ etc., etc., etc. (almost every management and scientific fields)

Introduction: modeling challenges

Model specification: to be able to define the "real" relationship between variables x and y .

For example, the attendance to a music festival may increase with temperature forecast for moderate temperature range and then decrease for very high temperatures. A useful model should take that up-down effect of the weather.

Variables selection: to be able to discover and select useful x 's.

For example, the attendance may be explained by the price of the butter in the model but this is useless. On the other hand, it could be explained by the attendance to another festival happening just before. However, that information is not available.

Introduction: What is a **statistical** model?

A model built from the observation of a sample $\{y_i, x_i\}_{i=1}^n$ of individuals, cases, subjects, occurrences, etc.

Statistical challenge: partial observation of the population.

For example, the model aims at explaining the expenses of a customer in a supermarket according to its characteristics.

However, only 100 people (the sample) are in the study (in opposition to the whole population).

For example, the model aims at predicting the demand in the future for a given product. However, only past demands can be observed (of course...).

Statistical results aim at being extrapolated to the whole (unobserved) population; a difficult task.

Some reminders of univariate statistics

In statistic, observed data constitute the sample. Unobserved data constitute the population (of interest).

Most used data type:

- ▶ Numerical:
 - ▶ Continuous: possible results can't be counted (e.g. time to event, height, spending, etc.)
 - ▶ Discrete: possible results can be counted (e.g. number of people waiting in line, calls to a call center, etc.)
- ▶ Categorical: possible results are categories
 - ▶ Nominal: no order (e.g. sex, spoken language, etc.)
 - ▶ Ordinal: ordered (e.g. level of education, level of satisfaction, etc.)

Some reminders of univariate statistics

Statistical notions for numerical observation

Sample	Population
Observation y_i	Random variable Y
Sample size n	Pop. size N , ∞ , or continuous
Mean $\bar{y} = \frac{1}{n} \sum_i y_i$	Expectation $E(Y)$
Variance $s^2 = \frac{1}{n-1} \sum_i (y_i - \bar{y})^2$	$Var(Y) = E((Y - E(Y))^2)$
Standard deviation $s = \sqrt{s^2}$	$SD(Y) = \sqrt{Var(Y)}$

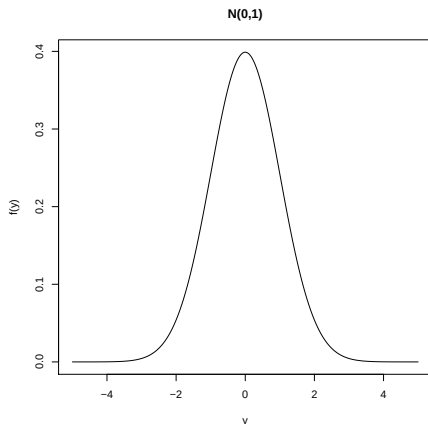
Object describing the distribution

Type	Sample	Population
Num. continuous	Histogram	Density function
Num. discrete	Probability table	Probability function
Categorical	Probability table	Probability function

Some reminders of univariate statistics

Standard normal distribution $Y \sim N(0, 1)$

$$f(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-y^2/2\right), \quad E(Y) = 0, \quad \text{Var}(Y) = 1$$



The 68 – 95 – 99.7 rule:

- ▶ $P(-1 \leq Y \leq 1) = 0.68$
- ▶ $P(-2 \leq Y \leq 2) = 0.95$
- ▶ $P(-3 \leq Y \leq 3) = 0.997$

Some reminders of univariate statistics

Normal distribution $Y \sim N(\mu, \sigma^2)$

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}, \quad E(Y) = \mu, \quad \text{Var}(Y) = \sigma^2$$

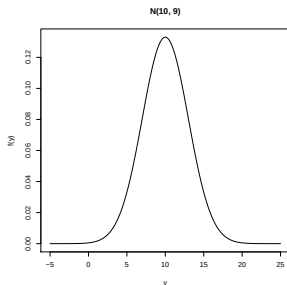
Link between $N(0, 1)$ and $N(\mu, \sigma^2)$:

$$Y = \mu + \sigma Z, \quad Z = \frac{Y - \mu}{\sigma}$$

where $Z \sim N(0, 1)$.

The 68 – 95 – 99.7 rule:

- ▶ $P(-\sigma \leq Y \leq \sigma) = 0.68$
- ▶ $P(-2\sigma \leq Y \leq 2\sigma) = 0.95$
- ▶ $P(-3\sigma \leq Y \leq 3\sigma) = 0.997$



Some reminders of univariate statistics

For $Z \sim N(0, 1)$:

$$P(Z \leq q) = p$$

Computation using R,

- ▶ Density: for given x , $f(x)$ is `dnorm(x, mean=0, sd=1)`.
- ▶ Probability: for given q , p =`pnorm(q, mean=0, sd=1)`.
- ▶ Quantile: for given p , q =`qnorm(p, mean=0, sd=1)`.
- ▶ Common notation of the quantile: z_p .

Some reminders of univariate statistics

Student t distribution with df degrees of freedom $Y \sim t_{df}$.

$$E(Y) = 0, \quad \text{Var}(Y) = \frac{df}{df - 2}.$$

Density function is complex.

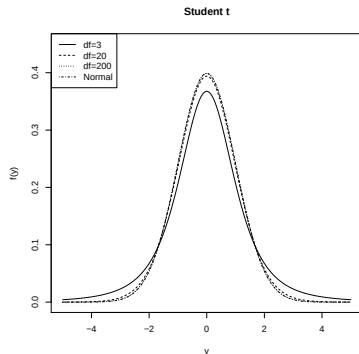
In R, Density: `dt(x, df)`.

Probability: for $t_{df}(q)$, use `pt(q, df)`

Quantile: for $t_{df;p}$, use `qt(p, df)`

Approximation by the normal:

When $df \rightarrow \infty$, t_{df} tends to $N(0, 1)$.



Some reminders of univariate statistics

Chi-square χ^2 distribution with df degrees of freedom $Y \sim \chi^2_{df}$.

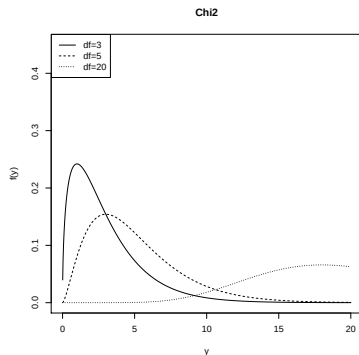
$$E(Y) = df, \quad \text{Var}(Y) = df^2$$

Density function is complex.

In R, Density: `dchisq(x, df)`.

Probability: for $\chi^2_{df}(q)$, use `pchisq(q, df)`

Quantile: for $\chi^2_{df;p}$, use `qchisq(p, df)`



Some reminders of univariate statistics

Fisher F distribution with df_1 and df_2 degrees of freedom

$$Y \sim F_{df_1, df_2}.$$

$$E(Y) = \frac{df_2}{df_2 - 2},$$

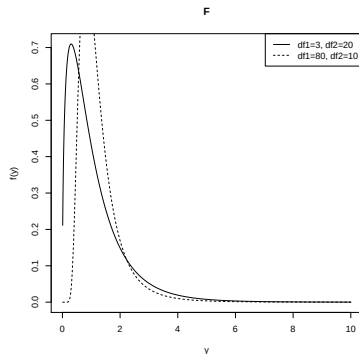
$$Var(Y) = \frac{2df_2^2(df_1 + df_2 - 2)}{df_1(df_2 - 2)^2(df_2 - 4)}$$

Density function is complex.

In R, Density: `df(x, df1, df2)`.

Probability: for $F_{df_1, df_2}(q)$, use `pf(q, df1, df2)`

Quantile: for $F_{df_1, df_2; p}$, use `qf(p, df1, df2)`



Some reminders of univariate statistics

Some links between these distributions:

- ▶ If $Y \sim N(0, 1)$ then $Y^2 \sim \chi_1^2$.
- ▶ If $Y_1, \dots, Y_n \sim N(0, 1)$, independent, then $\sum_{i=1}^n Y_i^2 \sim \chi_n^2$
- ▶ If $Y \sim N(0, 1)$ and $V \sim \chi_v^2$, independent, then

$$\frac{Y}{\sqrt{V/v}} \sim t_v$$

- ▶ If $V \sim \chi_v^2$ and $W \sim \chi_w^2$, independent, then

$$\frac{V/v}{W/w} \sim F_{v,w}.$$

Some reminders of univariate statistics

Point estimate of the mean (sample estimate): $\bar{y}$

Confidence interval at level $1 - \alpha$ (population estimate):

$$\left[\bar{y} - t_{n-1;1-\alpha/2} s / \sqrt{n}, \bar{y} + t_{n-1;1-\alpha/2} s / \sqrt{n} \right]$$

Approximate (normal) confidence interval:

$$\left[\bar{y} - z_{1-\alpha/2} s / \sqrt{n}, \bar{y} + z_{1-\alpha/2} s / \sqrt{n} \right]$$

Valid if n is large.

α	$z_{1-\alpha/2}$
0.40	0.84
0.30	1.04
0.20	1.28
0.10	1.64
0.05	1.96

In R, $z_{1-\alpha/2}$ is `qnorm(p=1-a/2)`.

Some reminders of univariate statistics

Bilateral t-test (two-sided):

$$H_0 : E(Y) = m_0 \quad \text{versus} \quad H_1 : E(Y) \neq m_0$$

Statistic:

$$t = \frac{\bar{y} - m_0}{s/\sqrt{n}}$$

Distribution if H_0 is true: t_{n-1} (Student with $\text{df}=n-1$).

p -value:

$$p = 2 - 2 \times t_{n-1}(t).$$

Test at level α : reject H_0 if (equiv.)

- ▶ $|t| > t_{n-1;1-\alpha/2}$,
- ▶ $p < \alpha$.

Some reminders of univariate statistics

Bilateral t-test (one-sided):

$$H_0 : E(Y) = m_0 \quad \text{versus} \quad H_1 : E(Y) > m_0 \quad [<]$$

Statistic:

$$t = \frac{\bar{y} - m_0}{s/\sqrt{n}}$$

Distribution if H_0 is true: t_{n-1}

p -value:

$$p = 1 - t_{n-1}(t) \quad [p = t_{n-1}(t)]$$

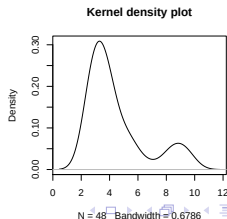
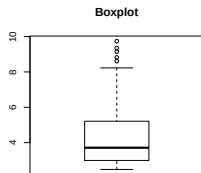
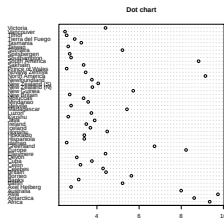
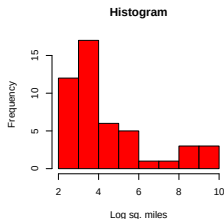
Test at level α : reject H_0 if (equiv.)

- ▶ $t > t_{n-1;1-\alpha}, \quad [t < t_{n-1;\alpha}]$
- ▶ $p < \alpha.$

Some reminders of univariate statistics

Graphical representation of numerical data: histogram, dot chart, boxplot, kernel density plot

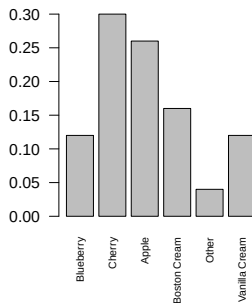
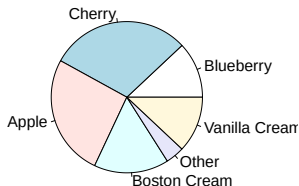
Example: world's areas ($\times 1000 \text{ mi}^2$; log) of the landmasses exceeding $10,000 \text{ mi}^2$.



Some reminders of univariate statistics

Graphical representation of categorical data: pie chart and bar plot

Example: Pie chart of pie sales.



Description of the dependence between two variables

Context: same subject i , two measures x_i, y_i .

How to describe their dependence?

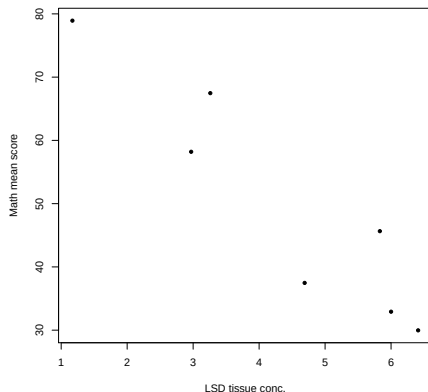
For examples:

- ▶ Height and weight of a subject: high/low together?
- ▶ Speed and consumption along a journey: high/low together?
- ▶ Happiness and productivity, gender and salary, age and

Dependence of two numerical variables: scatter plot

Graph of x versus y .

Example: LSD concentration in tissue versus a Math exam score



Dependence is negative and (approximately) linear.

Source: Wagner, Agahajanian, and Bing (1968). Correlation of Performance Test Scores with Tissue Concentration of Lysergic Acid Diethylamide in Human Subjects. Clinical Pharmacology and Therapeutics, Vol.9 pp635-638.

Dependence between two numerical variables: correlation

Correlation measures the linear dependence between x and y unconditionally on the other variables

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

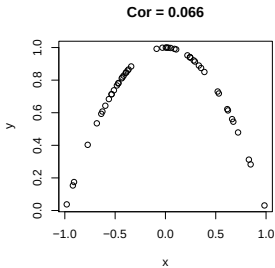
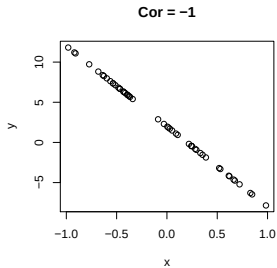
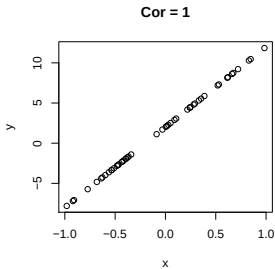
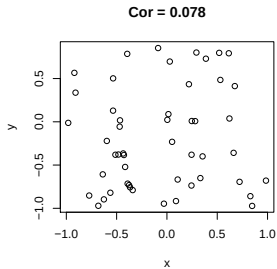
Correlation is between -1 (strong negative) and 1 (strong positive).

In the example: $r_{xy} = -0.94$

```
> cor(lsd)
```

	LSD	Math
LSD	1.0000000	-0.9369285
Math	-0.9369285	1.0000000

Correlation measures linear dependence



Population correlation and correlation test

The correlation of the population (unobserved):

$$\text{Cor}(X, Y) = \frac{E[(X - E(X))(Y - E(Y))]}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Test on the correlation

$$H_0 : \text{Cor}(X, Y) = 0 \quad \text{versus} \quad H_1 : \text{Cor}(X, Y) \neq 0$$

$$\text{Statistics under } H_0: t = \frac{r_{xy}}{\sqrt{(1-r_{xy}^2)/(n-2)}} \sim t_{n-2}$$

$$\text{Reject } H_0: |t| > t_{n-2; 1-\alpha/2}$$

$$\text{P-value: } p = 2 - 2 \times t_{n-2}(t)$$

Correlation test: example

```
> cor.test(x=lsd$LSD, y=lsd$Math)
```

Pearson's product-moment correlation

data: lsd\$LSD and lsd\$Math

t = -5.994, df = 5, p-value = 0.001854

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.9908681 -0.6244782

sample estimates:

cor

-0.9369285

H_0 is rejected at 5% since the p -value is lower than 5%.

Covariance

Covariance is a non standardized version of the correlation:

- ▶ Sample covariance:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- ▶ Population covariance:

$$\text{Cov}(X, Y) = E\{(X - E(X))(Y - E(Y))\}.$$

- ▶ Variance: the covariance of a variable with itself is the variance

$$s_{xx} = s_x^2, \quad \text{Cov}(X, X) = \text{Var}(X).$$

Example:

```
> cov(lsd)
      LSD      Math
LSD    3.745824 -33.74787
Math -33.747874 346.36389
> cor(lsd)[1,2]*sd(lsd$Math)*sd(lsd$LSD)
[1] -33.74787
```

Units

Suppose that X is in meters m , and $x_1, \dots, x_n$ is a sample. Then

- ▶ $E(X)$ and $\bar{x}$ are in m .
- ▶ $SD(X)$ and s_x are in m .
- ▶ $Var(X)$ and s_y^2 are in m^2 .

Furthermore, suppose that another measure is taken, say Y is in g , and $y_1, \dots, y_n$ the sample, then

- ▶ $Cov(X, Y)$ and s_{xy} are in $m \times g$.
- ▶ $Cor(X, Y)$ and r_{xy} have no units.

Notations

Frequent notations use Greek letters:

- ▶ $E(X)$: μ_X or μ (if mention to X is not needed),
- ▶ $Var(X)$: σ_X^2 or σ^2 ,
- ▶ $SD(X)$: σ_X or σ ,
- ▶ $Cov(X, Y)$: σ_{XY} ,
- ▶ $Cor(X, Y)$: ρ_{XY} or ρ .

Covariance and correlation matrices

Extending to more than two variables, the covariance matrix is the matrix of covariances:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}, \quad S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

where

$$s_{jj'} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ij'} - \bar{x}_{j'}), \quad \sigma_{jj'} = E\{(X_j - E(X_j))(X_{j'} - E(X_{j'}))\}.$$

The diagonal elements of Σ and S are the variances (population and sample) of the variables

$$\sigma_{jj} = \text{Var}(X_j), \quad s_{jj} = s_j^2.$$

Covariance and correlation matrices

Example: Macroeconomic data set `longley` (see `?longley` for details).

```
> signif(cov(longley), digit=4)
```

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year	Employed
GNP.deflator	116.50	1064.0	625.9	349.0	73.50	50.92	36.80
GNP	1064.00	9879.0	5612.0	3088.0	685.20	471.00	343.30
Unemployed	625.90	5612.0	8732.0	-1154.0	446.30	297.30	164.90
Armed.Forces	349.00	3088.0	-1154.0	4843.0	176.40	138.20	111.80
Population	73.50	685.2	446.3	176.4	48.39	32.92	23.46
Year	50.92	471.0	297.3	138.2	32.92	22.67	16.24
Employed	36.80	343.3	164.9	111.8	23.46	16.24	12.33


```
> with(longley, var(GNP.deflator))  
[1] 116.4576
```

Covariance and correlation matrices

Correlation matrices are similar:

$$P = \begin{bmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1p} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p1} & \rho_{p2} & \cdots & \rho_{pp} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix}$$

where

$$r_{jj'} = \frac{s_{jj'}}{s_j s_{j'}}, \quad \rho_{jj'} = \frac{\sigma_{jj'}}{\sigma_j \sigma_{j'}}.$$

The diagonal elements of P and R are equal to 1.

Covariance and correlation matrices

Example:

```
> signif(cor(longley), digit=4)
```

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year	Employed
GNP.deflator	1.0000	0.9916	0.6206	0.4647	0.9792	0.9911	0.9709
GNP	0.9916	1.0000	0.6043	0.4464	0.9911	0.9953	0.9836
Unemployed	0.6206	0.6043	1.0000	-0.1774	0.6866	0.6683	0.5025
Armed.Forces	0.4647	0.4464	-0.1774	1.0000	0.3644	0.4172	0.4573
Population	0.9792	0.9911	0.6866	0.3644	1.0000	0.9940	0.9604
Year	0.9911	0.9953	0.6683	0.4172	0.9940	1.0000	0.9713
Employed	0.9709	0.9836	0.5025	0.4573	0.9604	0.9713	1.0000

Correlations, tests, and scatter plots

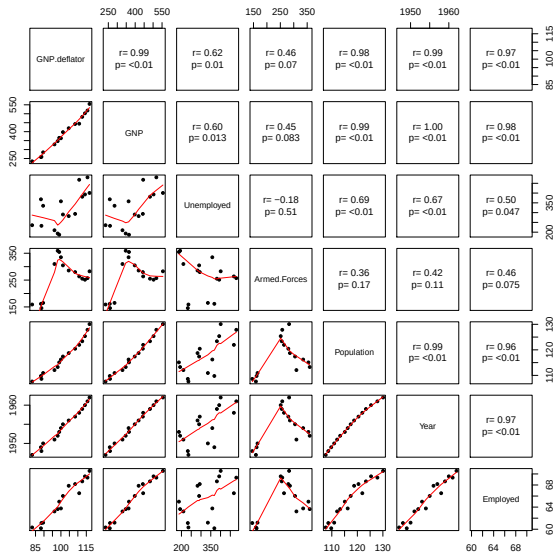
Combining these two tools¹ in R.

```
panel.cor <- function(x, y, digits = 2, cex.cor, ...)  
{  
  usr <- par("usr"); on.exit(par(usr))  
  par(usr = c(0, 1, 0, 1))  
  # correlation coefficient  
  r <- cor(x, y)  
  txt <- format(c(r, 0.123456789), digits = digits)[1]  
  txt <- paste("r= ", txt, sep = "")  
  text(0.5, 0.6, txt)  
  
  # p-value calculation  
  p <- cor.test(x, y)$p.value  
  txt2 <- format(c(p, 0.123456789), digits = digits)[1]  
  txt2 <- paste("p= ", txt2, sep = "")  
  if(p<0.01) txt2 <- paste("p= ", "<0.01", sep = "")  
  text(0.5, 0.4, txt2)  
}  
  
> pairs(longley, upper.panel = panel.cor, pch=20, lower.panel=panel.smooth)
```

\$

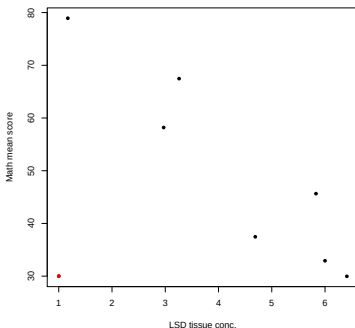
¹Source: <https://www.r-bloggers.com/scatter-plot-matrices-in-r/>

Correlations, tests, and scatter plots



(Un-)Robustness

The correlation is influenced by extreme values. A low math score, though no drug. The correlation goes down to -0.52 .



Several correlations exist: Pearson's correlation, Spearman's rho, Kendall's tau.

Spearman's ρ

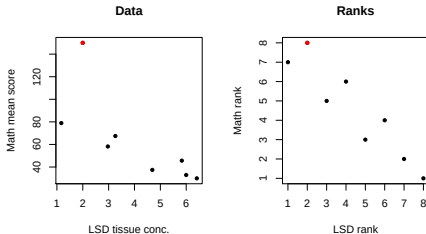
Instead of using the correlation of the observed values $(y_i, x_i)_{i=1}^n$, the correlation of the ranks is computed: an attempt to be more robust against extremes.

Example: LSD data.

LSD	LSD rank	Math rank	Math
1.17	1	7	78.93
2.00	2	8	150.00
2.97	3	5	58.20
3.26	4	6	67.47
4.69	5	3	37.47
5.83	6	4	45.65
6.00	7	2	32.92
6.41	8	1	29.97

Spearman's ρ

	Without extreme value	With extreme value
Pearson	-0.94	-0.75
Spearman	-0.93	-0.93



Spearman's ρ test

Similar to the correlation test

$$H_0 : \rho_S(X, Y) = 0 \quad \text{versus} \quad H_1 : \rho_S(X, Y) \neq 0$$

Statistics under H_0 : $t = r_S \sqrt{\frac{n-2}{1-r_S^2}} \sim t_{n-2}$.

Reject H_0 : $|t| > t_{n-2; 1-\alpha/2}$

P-value: $p = 2 - 2 \times t_{n-2}(t)$

Spearman's ρ test

```
> cor.test(x=lsd2$LSD, y=lsd2$Math, method="spearman")
```

Spearman's rank correlation rho

data: lsd2\$LSD and lsd2\$Math

S = 162, p-value = 0.002232

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

-0.9285714

H_0 is rejected at 5% since the p -value is lower than 5%.

Kendall's τ

Yet another rank based correlation is the Kendall's tau. The principle is more complex than the two others.

- ▶ Compute the number of concordant pairs C and discordant pairs D in both samples of ranks.
- ▶ If there is no relation between X and Y , the difference between these should be small.
- ▶ If the number of concordant pairs is large, then there is a positive relationship.
- ▶ If the number of discordant pairs is large, then there is a negative relationship.

The statistics is defined as

$$r_{\tau} = \frac{C - D}{n(n - 1)/2}$$

Kendall's τ

Example:

LSD	LSD rank	Math rank	Math
1.17	1	7	78.93
2.00	2	8	150.00
2.97	3	5	58.20
3.26	4	6	67.47
4.69	5	3	37.47
5.83	6	4	45.65
6.00	7	2	32.92
6.41	8	1	29.97

The pair LSD:(1, 2) and Math:(7, 8) is concordant (both increasing). The pair LSD:(1, 3) and Math:(7, 5) is discordant (decreasing vs increasing). Etc.

In total: $C = 3$ and $D = 25$. Thus

$$r_{\tau} = (3 - 25)/(8 \times 7/2) = -0.786.$$

Kendall's τ : Why $n(n-1)/2$?

For n observations, the number of pairs is

$$n-1 + n-2 + \cdots + 2 + 1 = \frac{n(n-1)}{2}$$

Thus the statistics is bounded: $-1 \leq r_\tau \leq 1$.

Example:

Pairs including observation 1: 7

Pairs including observation 2: 6

Pairs including observation 3: 5

Pairs including observation 4: 4

Pairs including observation 5: 3

Pairs including observation 6: 2

Pairs including observation 7: 1

Total: $28 = 8 \times 7/2$

Kendall's τ test

Similar to the correlation test

$$H_0 : \rho_\tau(X, Y) = 0 \quad \text{versus} \quad H_1 : \rho_\tau(X, Y) \neq 0$$

Statistics under H_0 : $z = \frac{3(C-D)}{\sqrt{n(n-1)(2n+5)/2}} \stackrel{\sim}{\sim} N(0, 1)$.

Reject H_0 : $|z| > z_{1-\alpha/2}$

P-value: $p = 2 - 2 \times N(0, 1)(t)$.

Kendall's τ test

```
> cor.test(x=lsd2$LSD, y=lsd2$Math, method="kendall")
```

Kendall's rank correlation tau

data: lsd2\$LSD and lsd2\$Math

T = 3, p-value = 0.005506

alternative hypothesis: true tau is not equal to 0

sample estimates:

tau

-0.7857143

H_0 is rejected at 5% since the p -value is lower than 5%.

Measure of dependence: discussion

- ▶ Spearman's ρ and Kendall's τ are more robust to extreme values than Pearson's correlation.
- ▶ Pearson's correlation is more often use because of its convenient properties with normally distributed data.
- ▶ Dependence measure helps to measure dependence and to compare dependence patterns.
- ▶ They are on different scales and should not be compared: e.g. do not use the Spearman's ρ as a replacement of the Pearson's correlation unless you really know what you are doing.
- ▶ All dependence measures provide only a partial view on the dependence.

Normality and Pearson's correlation

It looks like rank based measures are better. So why is the correlation so often used?

When the variables x and y are both normally distributed then the correlation fully describes their dependence structure. In this case, the Pearson's correlation is the best possible description.

Note that:

- ▶ This is why correlation is so often used.
- ▶ This implies that there is no extreme value or that they should be put aside in some way.

Statistical modeling

Module 02

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Autumn 2018

Further than correlation and dependence measures

Correlations and dependence measures

- ▶ Quantify the dependence.
- ▶ Do not qualify the type of dependence.
- ▶ Are limited to pairwise dependence.
- ▶ Are not useful for prediction.

We now build models



where

- ▶ the dependence of y on x is explicit and can be used for prediction,
- ▶ x can be several variables, $x_1, \dots, x_p$,
- ▶ the uncertainty is quantified.

Variables in the study

There are two types of variables:

- ▶ Variables $x_1, \dots, x_p$, called features (machine learning context), independent variables, factors or covariates¹.
- ▶ A response y , also called outcome or dependent variable.

The response variable has a central role. It is a variable of interest either

- ▶ to be predicted.
- ▶ to study the relation between y and each features in x .

In the following, we use x for all the features, and detail $x_1, \dots, x_p$ when needed.

¹and other names in other scientific fields...

Examples

Interested in answering the above research question, (Willerman, et al, 1991) studied a sample of $n = 38$ college students²:

- ▶ Response (y): Performance IQ scores (PIQ) from the revised Wechsler Adult Intelligence Scale. This variable served as the investigator's measure of the individual's intelligence.
- ▶ Potential predictor (x_1): Brain size based on the count obtained from MRI scans (given as count/10'000).
- ▶ Potential predictor (x_2): Height in inches.
- ▶ Potential predictor (x_3): Weight in pounds.

²source: <https://onlinecourses.science.psu.edu/stat501/node/284>

Examples

(Colby, et al, 1987) wanted to find out if swallows, which live in underground burrows, also alter how they breathe. In an underground burrow, they varied the percentage of oxygen at four different levels (13%, 15%, 17%, 19%) and the percentage of carbon dioxide at five different levels (0%, 3%, 4.5%, 6%, 9%). Under each of the resulting $5 \times 4 = 20$ experimental conditions, the researchers observed the total volume of air breathed per minute for each of 6 nestling bank swallows. They obtained a sample of $n = 120$ nestling bank swallows³:

- ▶ Response (y): percentage increase in "minute ventilation," (Vent), i.e., total volume of air breathed per minute.
- ▶ Potential predictor (x1): percentage of oxygen (O2) in the air the baby birds breathe.
- ▶ Potential predictor (x2): percentage of carbon dioxide (CO2) in the air the baby birds breathe.

³source: <https://onlinecourses.science.psu.edu/stat501/node/285>

Examples

Example: In a company the weekly sales of ice-cream y is dependent on the advertising budget x_1 , the average temperature x_2 , and the holiday season x_3 . Past data are used to forecast the future sales in order to manage the supply/production processes.

Example: In a large professional association, a study about equality in employee salaries y is performed. The independent variables are the gender x_1 , the company sector x_2 , the education level x_3 , the seniority x_4 , the position in the company x_5 , and the hierarchy in the company x_6 . The impact of the gender on the salary (at same level of the other variables) is of interest to determine if anti-discrimination actions should be taken.

The model

To qualify/quantify the impact of x on y , a model is built

$$y = f(x_1, \dots, x_p) + e.$$

where

- ▶ function $f(x_1, \dots, x_p)$ represents what are the expected values y for given values of x .
- ▶ e is the error term of the model, expected to be 0. The error helps to describe the uncertainty of the model.

The function f needs to be estimated in order to give a good description of the outcome y for given x . The category of available functions is very large. The most used is linear functions, which gives the well known regression models which is the central topic of this course.

The data

The model is calibrated (estimated, more later) in order to match a sample of observations. Except specified differently, the notation is

- ▶ i denotes the observation label (e.g. subject, week, case, etc.).
- ▶ n is the sample size: number of observations.
- ▶ (y_i, x_i) or $(y_i, x_{i1}, \dots, x_{ip})$ the observed values (response and features) for observation i .

The observations are assumed to be independent given x .

Example: in the study on college students, i is a student label, $n = 38$, and for example, for subject 1, we have

$$y_1 = 124, x_{11} = 81.69, x_{12} = 64.5, x_{13} = 118.$$

Subjects 1 and 2 are independent: knowing y_1 does not bring any extra information on y_2 if x_2 is known.

Regression

Regression is a model where the function f is assumed to be a linear combination of the variables x .

Assuming that we have n observations $\{y_i, x_{i1}, \dots, x_{ip}\}$, then the model is

$$y_i = b_0 + b_1 x_{i1} + \dots + b_p x_{ip} + e_i,$$

where

- ▶ $b_0, b_1, \dots, b_p$ are the coefficients
- ▶ e_i is the residual error

$$e_i = y_i - (b_0 + b_1 x_{i1} + \dots + b_p x_{ip}).$$

Note: for simplicity, we dropped the dependence on b . To be "super precise" we could write $y(x; b)$ and $e(x; b)$.

Regression

Given a set of coefficients $b = (b_0, \dots, b_p)$, for a subject i having characteristics $\{y_i, x_{1i}, \dots, x_{pi}\}$, the prediction of y_i is

$$y(x_i) = b_0 + b_1x_{1i} + \dots + b_px_{pi}$$

The estimated error is $e(x_i) = y_i - y(x_i)$. It is expected to be 0.

Several names are given to these models:

- ▶ regression,
- ▶ linear regression,
- ▶ linear model, general linear model (not to be confounded to generalized linear models),
- ▶ multiple regression (as opposed to simple regression, because there are more than one x).

Regression: illustration

In a population of children, seasonal pocket money (y in \$) is studied as a function of the age (x_1 in yrs) and the average grand-parent age (x_2 in yrs).

Model:

$$y = 307.71 + 10.94x_1 + 2.85x_2 + e$$

i	y_i	x_{1i}	x_{2i}	$y(x_i)$	$e(x_i)$
1	565	10	50	559.6	5.39
2	601	13	60	620.9	-19.9
3	560	14	40	574.9	-14.9
...	...	...	...	...	...

Regression: interpretation of the coefficients

Coefficient b_j is the average increase of y for an increase of 1 unit of x_j . Sometimes they are called the sensitivity of y on x_j .

Example:

- ▶ Pocket money increases in average by \$10.94 when age increases by 1 year,
- ▶ Pocket money increases in average by \$2.85 when the average grand-parent age increases by 1 year.

Regression: out-of-sample prediction

New subject prediction: y is unknown, $x = (x_1, \dots, x_p)$ is given,

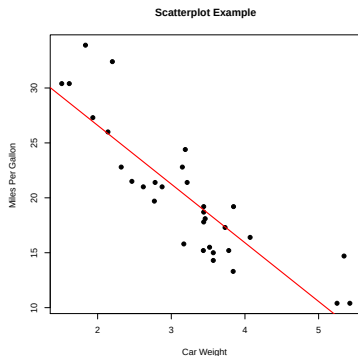
$$y(x) = b_0 + b_1x_1 + \dots + b_px_p$$

Example: for pocket money, a new 8 year-old child (y unknown) with 55 year-old grand-parents is predicted:

$$y = 307.71 + 10.94 \times 8 + 2.85 \times 55 = 551.98$$

Regression: graphical representation

For the simple regression (i.e. $p = 1$), use scatter plot and add the regression line (red). Below, a car consumption y (Miles per Gallon) as a function of its weight x_1 .



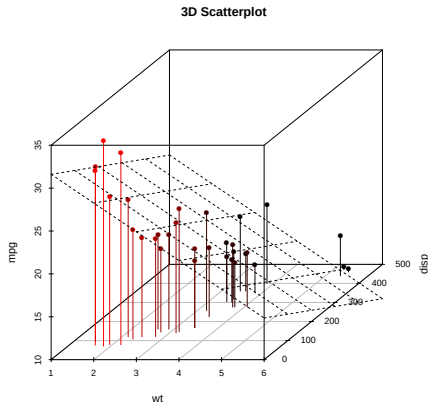
$$\text{Model: } y = 37.3 - 5.3x_1 + e$$

Regression: graphical representation

With $p = 2$, one could use a regression plan (not often in practice).

Below consumption y as function of the weight x_1 and the displacement x_2 .

$$y = 35.0 - 3.35x_1 - 0.018x_2 + e$$



Estimation of a regression: OLS

There are thus $p + 2$ parameters to estimate: $(b_0, \dots, b_p)$ and σ . From a sample $\{y_i, x_1, \dots, x_p\}_{i=1}^n$, what are the best coefficients $b = (b_0, \dots, b_p)$ to use?

Minimize the sum of squares:

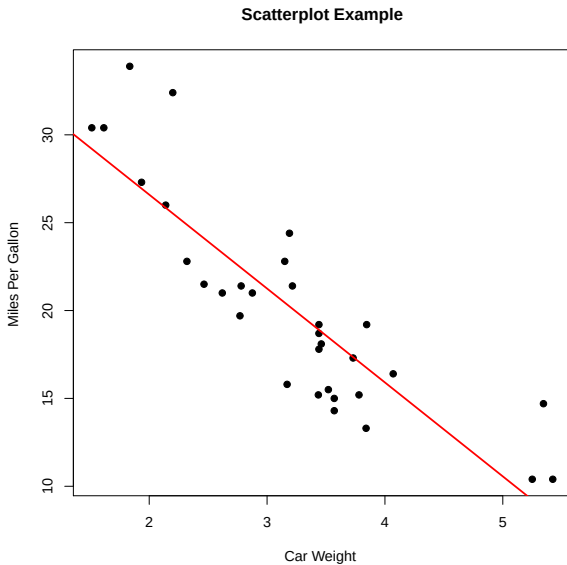
$$\begin{aligned} SS(b) &= \sum_{i=1}^n e(x_i)^2 = \sum_{i=1}^n \{y_i - y(x_i)\}^2 \\ &= \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip})^2 \end{aligned}$$

Let $\hat{b}$ be the solution,

$$\hat{b} = \arg \min_b SS(b)$$

This is the Ordinary Least Squares (OLS).

OLS: illustration



OLS: calculations

Compute the derivatives and equalize to 0:

$$\left. \frac{\partial SS(b)}{\partial b} \right|_{b=\hat{b}} = 0.$$

For b_0 :

$$\frac{\partial SS(b)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \cdots - b_p x_{ip}).$$

For $b_j, j = 1, \dots, p$:

$$\frac{\partial SS(b)}{\partial b_j} = -2 \sum_{i=1}^n x_{ij} (y_i - b_0 - b_1 x_{i1} - \cdots - b_p x_{ip}).$$

OLS: calculations

The estimating equations for b are

$$\sum_{i=1}^n x_{ij}(y_i - \hat{b}_0 - \hat{b}_1 x_{i1} - \cdots - \hat{b}_p x_{ip}) = 0, \quad j = 0, \dots, p.$$

where $x_{i0} = 1$ for all $i = 1, \dots, n$.

Finding $\hat{b}$ by solving these estimating equations can be done using iterative numerical algorithms. A convenient way of representing (and solving) the system is by matrix representation.

Matrix notations

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} b_0 \\ \vdots \\ b_p \end{pmatrix} + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix},$$

$$y = Xb + e$$

Sum of squares:

$$SS(b) = (y - Xb)^T (y - Xb) = e^T e = \|e\|^2$$

Estimating equations:

$$X^T (y - X\hat{b}) = 0$$

Ordinary Least Squares estimate:

$$\hat{b} = (X^T X)^{-1} X^T y$$

OLS: predictions and errors

OLS predictions:

$$\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_{i1} + \cdots + \hat{b}_p x_{ip},$$

OLS errors:

$$\hat{e}_i = y_i - \hat{y}_i.$$

In matrix notations:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{b}}, \quad \hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}}$$

Example

A real estate firm builds a selling price prediction model for a house (y in \$1'000) using Sq feet (x_1), Age (x_2), Bedrooms (x_3), Bathrooms (x_4), # cars in the garage (x_5). They use a data base with historical prices ($n = 250$). Using the OLS, they find

$$\text{Price} = 31.1 + 63.1x_1 - 1.14x_2 - 8.4x_3 + 3.5x_4 + 28.2x_5 + e$$

Could you explain the negative coefficient $\hat{b}_3$?

Example:

For the IQ study, when calling the R function `lm`, the OLS is computed and extracted when calling the `summary` function:

```
> mod <- lm(PIQ ~ Brain + Height + Weight, data=IQ)
> summary(mod)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

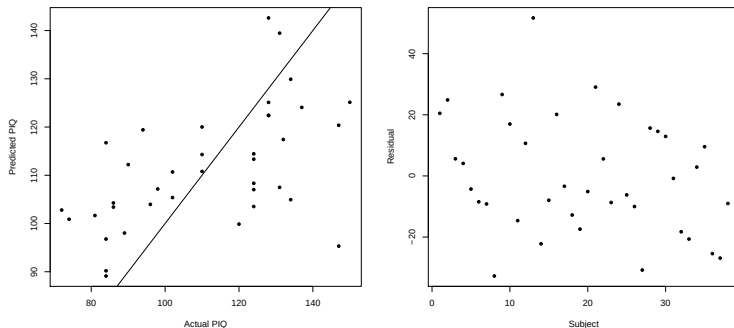
```
---
[...]
```

The fitted model is

$$y = 111.4 + 1.060x_1 - 2.732x_2 + 0.00056x_3 + e.$$

Example:

The (in-sample) predicted responses against the actual values, $\hat{y}$ vs y and the residuals ($\hat{e}$) are shown below



Conclusion: from the left-hand plot we see that the prediction is not very good.

Sum of squares

Sum of squares are important quantities to qualify the model quality (goodness-of-fit) and much more.

Residual Sum of squares (e stands for **errors**):

$$SS_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (y - \hat{y})^T (y - \hat{y}) = \hat{e}^T \hat{e}.$$

Total Sum of squares:

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$$

Sum of squares of the **Regression**:

$$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Sum of squares decomposition

It can be shown that

$$SS_T = SS_R + SS_e$$

Interpretation:

- ▶ A perfect regression predict y very well, i.e. $y_i \approx \hat{y}_i$ and thus

$$SS_e \approx 0, \quad SS_T \approx SS_R.$$

- ▶ A large SS_R indicates a good model fit to the data,
- ▶ A large SS_e indicates a bad model fit to the data.

Coefficient of determination: R^2

It is an indicator of the model quality. It is defined as the proportion of variation in y explained by the variation of x

$$R^2 = \frac{SS_R}{SS_T} = \frac{SS_T - SS_e}{SS_T} = 1 - \frac{SS_e}{SS_T}$$

- ▶ Because $0 \leq SS_R \leq SS_T$, $0 < R^2 < 1$,
- ▶ A large* R^2 indicates a good model fit to the data.

*How much large? please don't ask...

R^2 : Example

In the IQ example,

```
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.79 on 34 degrees of freedom
Multiple R-squared: 0.2949, Adjusted R-squared: 0.2327
F-statistic: 4.741 on 3 and 34 DF, p-value: 0.007215

The R^2 is equal to 0.295.

R^2 : Example

In the real estate example, they found

$$SS_T = 1.27 \times 10^6, SS_e = 2.34 \times 10^5, SS_R = 1.04 \times 10^6.$$

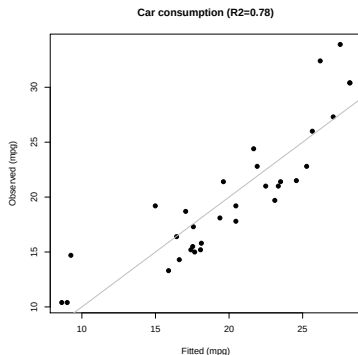
thus

$$R^2 = 1.04/1.27 = 0.82.$$

Thus, 82% of the variation of the price is explained by the variation of x_1, x_2, x_3, x_4, x_5 .

Graphical inspection of the model quality

The coefficient of determination R^2 is an indicator of model adequacy with the data. Often used in complement to the scatter plot of y against $\hat{y}$ (below for the car consumption example):



A good fit = points should be close to the (0,1) line (in grey)

Adjusted R^2

The adjusted R^2 is an alternative to the R^2 that accounts for the number of parameters in the model,

$$R_a^2 = 1 - \frac{SS_e/(n - p - 1)}{SS_T/(n - 1)}.$$

Suppose that two linear models achieve the same sum of squares SS_T and SS_e , and thus the same R^2 , then the model with the least number of parameters (p smaller) will have a higher R_a^2 .

The adjusted R^2 penalizes more complex models (more parameters). This principle is central in model selection (more later).

R^2 : Example

In the IQ example,

```
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.79 on 34 degrees of freedom
Multiple R-squared: 0.2949, Adjusted R-squared: 0.2327
F-statistic: 4.741 on 3 and 34 DF, p-value: 0.007215

The adjusted R^2 is equal to 0.2327. This is still bad...

R^2 : Example

In the real estate example, they found

$$SS_T = 1.27 \times 10^6, \quad SS_e = 2.34 \times 10^5, \quad SS_R = 1.04 \times 10^6.$$

thus

$$R_a^2 = 1 - \frac{2.34/(250 - 5 - 1)}{12.7/(250 - 1)} = 0.812.$$

We see that, when $p \ll n$ (here $5 \ll 250$), the R^2 and the adjusted R^2 are very close.

Statistical modeling

Module 03

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Bachelor, GSEM

Autumn 2018

Categorical variables

So far, we only have seen examples where the independent variables x are numerical.

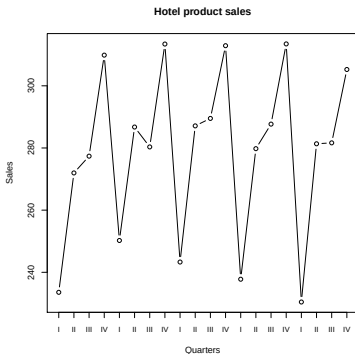
What if one or more of the x_j 's is categorical?

- ▶ Is it still a linear model?
- ▶ How can we write/fit the model?
- ▶ How can we make the prediction?

In fact, the model can be written in the same way. The procedure is first to create so called "dummy variables" and to use them as numerical variables.

Example

Consider the quarterly sales of a hotel product shown below. We ignore trend for simplicity.



The independent variable x is categorical with 4 levels: Quarter (I-IV).

Dummy variables

With 4 levels, we associate 3 dummy variables (nb of levels minus 1). They are defined as indicator whether the observation is in the corresponding quarter:

$$\delta_{II}(x) = \begin{cases} 1, & \text{if } x = II \\ 0, & \text{otherwise.} \end{cases}$$

$$\delta_{III}(x) = \begin{cases} 1, & \text{if } x = III \\ 0, & \text{otherwise.} \end{cases}$$

$$\delta_{IV}(x) = \begin{cases} 1, & \text{if } x = IV \\ 0, & \text{otherwise.} \end{cases}$$

Linear model with dummy variables

The model of the sales y at quarter x is

$$y = \beta_0 + \beta_{II}\delta_{II}(x) + \beta_{III}\delta_{III}(x) + \beta_{IV}\delta_{IV}(x) + e$$

Therefore, the prediction of the price in each quarter are

- ▶ Quarter I: $y(x) = \beta_0$
- ▶ Quarter II: $y(x) = \beta_0 + \beta_{II}$
- ▶ Quarter III: $y(x) = \beta_0 + \beta_{III}$
- ▶ Quarter IV: $y(x) = \beta_0 + \beta_{IV}$

Here Quarter I is the reference level. It is purely arbitrary and any level could be chosen as the reference.

Contrasts

The interpretation of the coefficients are in terms of difference to the reference level. These differences are called contrasts:

- ▶ β_0 : the average sales in Quarter I.
- ▶ β_{II} : contrast between Quarter II and Quarter I.
- ▶ β_{III} : contrast between Quarter III and Quarter I.
- ▶ β_{IV} : contrast between Quarter IV and Quarter I.

How would you estimate the contrast between Quarter II and Quarter III?

Example

R (and any statistical computer program) builds automatically the dummy variables and provides the estimation as usual.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	239.070	2.637	90.67	< 2e-16	***
QuarterII	42.320	3.729	11.35	4.59e-09	***
QuarterIII	44.240	3.729	11.87	2.42e-09	***
QuarterIV	71.920	3.729	19.29	1.67e-12	***

Residual standard error: 5.896 on 16 degrees of freedom

Multiple R-squared: 0.9596, Adjusted R-squared: 0.952

F-statistic: 126.7 on 3 and 16 DF, p-value: 2.328e-11

In Excel, you need to build the dummy variables by hand.

Dummy variables: general case

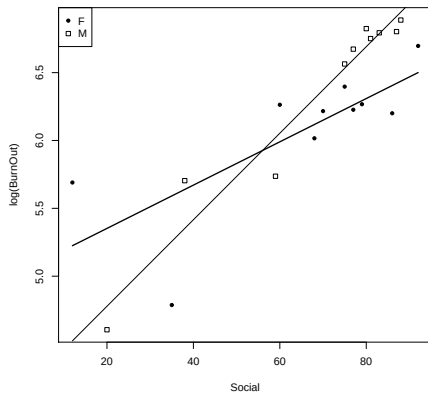
A categorical variable x with m levels $L_1, \dots, L_m$ is coded by $m - 1$ dummy variables

$$\delta_{L_k}(x_2) = \begin{cases} 1, & \text{if } x_2 = L_k, \\ 0, & \text{if } x_2 \neq L_k, \end{cases} \quad k = 2, \dots, m.$$

The level L_1 is the reference level. It has no dummy because the coefficients of the dummies are the contrasts with this level.

Mixing categorical and numerical variables

Example: In a study of the log-burn-out index (y) as a function of the sex (M/F) and a socialization index (x_1). Sex is a categorical variable. Consider two linear models:



Females

$$y = 5.033 + 0.016x_1 + e$$

Males

$$y = 4.14 + 0.031x_1 + e$$

Can we make one regression including the 2 factors Sex and Socialization Index?

Mixing categorical and numerical variables

Let x_2 be the sex with value M or F. The associated dummy variable is defined as

$$\delta_M(x_2) = \begin{cases} 0, & \text{if } x_2 = F \\ 1, & \text{if } x_2 = M. \end{cases}$$

The two models can be written with one single equation:

$$y = 5.033 - 0.893 \times \delta_M(x_2) + \{0.016 + 0.015 \times \delta_M(x_2)\}x_1 + e$$

Dummy variables

Indeed, if $x_2 = M$ then $\delta_M(x_2) = 1$ and

$$\begin{aligned}y &= 5.033 - 0.893 + \{0.016 + 0.015\}x_1 + e \\ &= 4.14 + 0.31x_1 + e\end{aligned}$$

And if $x_2 = M$ then $\delta_M(x_2) = 1$ and

$$y = 5.033 + 0.016x_1 + e$$

Example In R, the output is

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.033075	0.273818	18.381	3.5e-12	***
Social	0.015950	0.003946	4.042	0.000945	***
SexM	-0.892550	0.409303	-2.181	0.044482	*
Social:SexM	0.014934	0.005775	2.759	0.013976	*

Interpretations

The average difference between a female and a male with the same socialization index value, say x_1 , is

$$y_M - y_F = -0.893 + 0.015x_1$$

Thus, the two coefficients associated with the dummy $\delta_M(x_2)$ are the differences between F and M in

- ▶ The contrast in the intercepts
- ▶ The contrast in the slopes

Note that here Female is the reference level: the contrasts are associated with the level Male, the difference is "Male - Female". The reference is arbitrary can be modified by the user.

Matrix notations

Let

$$y = Xb + e$$

where the row X_i is

$$X_i = \left[1 \mid 0 \mid x_{i1} \mid 0 \right], \text{ if } i \text{ is a Female } (x_{i2} = F),$$

and

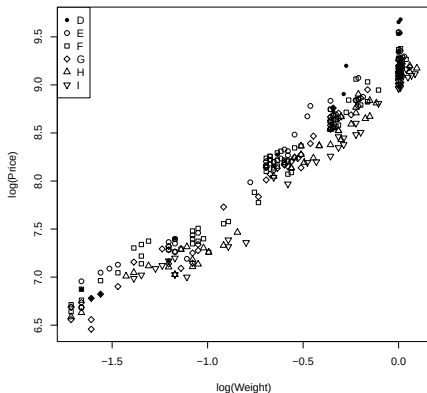
$$X_i = \left[1 \mid 1 \mid x_{i1} \mid x_{i1} \right] \text{ if } i \text{ is a Male } (x_{i2} = M),$$

Overall,

$$X_i = \left[1 \mid \delta_M(x_{i2}) \mid x_{i1} \mid x_{i1}\delta_M(x_{i2}) \right]$$

More than 2 levels

Example: Pricing diamond stones. The response is log-Price (y) and the independent variables are the log-weight (x_1), and the color (x_2). Color is categorical with 6 levels ($D - I$).



More than 2 levels

The analysis in R gives the coefficients:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	9.46974	0.04443	213.157	< 2e-16	***
log(Weight)	1.71716	0.05167	33.234	< 2e-16	***
ColorE	-0.21506	0.05334	-4.032	7.04e-05	***
ColorF	-0.29782	0.04977	-5.984	6.27e-09	***
ColorG	-0.34163	0.05083	-6.721	9.29e-11	***
ColorH	-0.41321	0.05014	-8.241	5.58e-15	***
ColorI	-0.50868	0.05156	-9.867	< 2e-16	***
log(Weight):ColorE	-0.17500	0.06478	-2.702	0.00730	**
log(Weight):ColorF	-0.19427	0.05860	-3.315	0.00103	**
log(Weight):ColorG	-0.17525	0.05884	-2.978	0.00314	**
log(Weight):ColorH	-0.12726	0.06189	-2.056	0.04062	*
log(Weight):ColorI	-0.13292	0.06589	-2.017	0.04456	*

Note: Color D is the reference level.

More than 2 levels

There are 5 dummy variables: no dummy for D (the reference).

$$\delta_E(x_2) = \begin{cases} 1, & \text{if } x_2 = E, \\ 0, & \text{if } x_2 \neq E, \end{cases}$$

$$\delta_F(x_2) = \begin{cases} 1, & \text{if } x_2 = F, \\ 0, & \text{if } x_2 \neq F, \end{cases}$$

$$\delta_G(x_2) = \begin{cases} 1, & \text{if } x_2 = G, \\ 0, & \text{if } x_2 \neq G, \end{cases}$$

$$\delta_H(x_2) = \begin{cases} 1, & \text{if } x_2 = H, \\ 0, & \text{if } x_2 \neq H, \end{cases}$$

$$\delta_I(x_2) = \begin{cases} 1, & \text{if } x_2 = I, \\ 0, & \text{if } x_2 \neq I. \end{cases}$$

There are $(\# \text{levels} - 1)$ dummy variables for a categorical factor.

Interactions between numeric and categorical variables

In the diamond example,

- ▶ Values associated to colors are called contrast. There is no contrast for the reference level.
- ▶ Values combining two factor levels are called interactions. They represent the variation of the one factor effect when the other factor level changes.

The interaction between a categorical factor and a numeric factor is the product between the dummy variables and x .

Interactions between numerical and categorical variables

Example: with diamond data

- ▶ Intercept (9.47) is the intercept for color D.
- ▶ Color E (-0.215) is the contrast of the intercept for color E to D: intercept for color E is $9.47 - 0.215$.
- ▶ $\log(\text{Weight})$ (1.72) is the slope for color D.
- ▶ $\log(\text{Weight}) : \text{ColorE}$ (-0.175) is the slope contrast for color E to D: the slope for color E is $1.72 - 0.175$. It is one interaction term.

Interactions between numerical and categorical variables

Example: with burn-out case

$$\begin{aligned} y = & 291.7 - 142.1\delta_M(x_2) - 4.61x_1 + 0.21x_1\delta_M(x_2) \\ & + 0.10x_1^2 + 0.06x_1^2\delta_M(x_2) + e, \end{aligned}$$

- ▶ -4.61 is the interaction between the sex and the linear term
- ▶ 0.06 is the interaction between the sex and the quadratic term.

Note that -142.1 could formally be seen as the interaction between the sex and the constant term¹.

¹nobody does that...

Interactions between two numerical variables

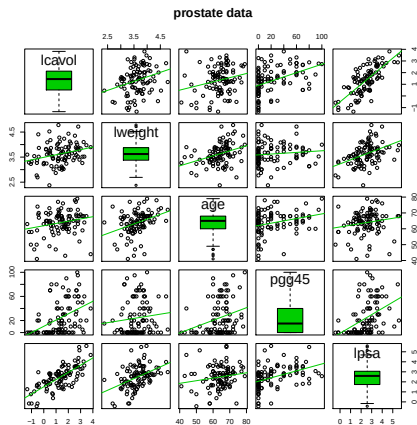
The interaction between a categorical factor and a numeric factor is the product between the dummy variables and x .

In the same way, the interaction between two numerical variables is also the product of the two variables.

Interactions between two numerical variables

Example:

The log-prostate specific antigen (lpsa) is inspected as a function of several variables (log-cancer volume, log-prostate weight, age, etc.).



Interactions between two numerical variables

Model with all predictors and an interaction between pgg45 and lpsa.

```
> mod <- lm(lcavol~lweight+age+pgg45*lpsa, data=prostate)
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.849617	0.864081	-0.983	0.328
lweight	-0.190851	0.224882	-0.849	0.398
age	0.014820	0.012172	1.218	0.227
pgg45	0.008882	0.009054	0.981	0.329
lpsa	0.739103	0.105066	7.035	3.62e-10 ***
pgg45:lpsa	-0.001386	0.003011	-0.460	0.646

```
---
[...]
```

Model equation

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_{34}x_3x_4 + e.$$

Estimated to

$$y = -0.850 - 0.191x_1 + 0.015x_2 + 0.009x_3 + 0.739x_4 - 0.0014x_3x_4 + e.$$

Interactions between two categorical variables

It is the product between the dummy variables of both factors.

Example: In diamond example, consider x_1 the color and a new variable x_2 a factor for the certified origin (YES/NO). Then, (below without the weight)

$$\begin{aligned} y = & b_0 + b_E \delta_E(x_1) + b_F \delta_F(x_1) + b_G \delta_G(x_1) + b_H \delta_H(x_1) + b_I \delta_I(x_1) \\ & + a_N \delta_N(x_2) \\ & + c_{EN} \delta_E(x_1) \delta_N(x_2) + c_{FN} \delta_F(x_1) \delta_N(x_2) + c_{GN} \delta_G(x_1) \delta_N(x_2) \\ & + c_{HN} \delta_H(x_1) \delta_N(x_2) + c_{IN} \delta_I(x_1) \delta_N(x_2) + e \end{aligned}$$

Interactions between two categorical variables

```
> fit2 <- lm(Price ~ Weight+Color*Certif, data=diamond2)
> summary(fit2)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2089.0	637.8	3.275	0.001182	**
Weight	11694.5	196.1	59.651	< 2e-16	***
ColorE	-3051.3	665.2	-4.587	6.65e-06	***
ColorF	-4137.3	639.6	-6.469	4.10e-10	***
ColorG	-4532.6	642.8	-7.052	1.26e-11	***
ColorH	-5103.2	635.0	-8.036	2.24e-14	***
ColorI	-5590.6	670.8	-8.334	3.00e-15	***
CertifYES	-2802.0	653.7	-4.286	2.46e-05	***
ColorE:CertifYES	1743.9	718.6	2.427	0.015831	*
ColorF:CertifYES	2672.6	687.4	3.888	0.000125	***
ColorG:CertifYES	2907.6	692.8	4.197	3.59e-05	***
ColorH:CertifYES	2964.0	689.2	4.300	2.32e-05	***
ColorI:CertifYES	2967.0	726.0	4.087	5.65e-05	***

```
---
[...]
```


Interactions between two categorical variables

- ▶ The number of model parameters is $1 + (g_1 - 1) + (g_2 - 1) + (g_1 - 1)(g_2 - 1)$, where g_j is the number of levels of factor j .
- ▶ This number is growing quickly with the number of levels.
- ▶ Interactions terms between categorical variables are often very difficult to interpret.

Higher order interactions

The interaction principle can be extended to higher order terms.
For example, two- and three way interactions,

```
> fit3 <- lm(Price ~ Weight*Color*Certif, data=diamond2)
```

```
> summary(fit3)
```

```
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	375.0	104154.5	0.004	0.997
Weight	13400.0	103635.0	0.129	0.897
ColorE	-5984.2	104161.6	-0.057	0.954
ColorF	-5058.4	104156.8	-0.049	0.961
ColorG	-4347.3	104158.0	-0.042	0.967
ColorH	-4493.6	104157.1	-0.043	0.966
ColorI	-4688.2	104164.4	-0.045	0.964
CertifYES	-3877.4	104155.4	-0.037	0.970
Weight:ColorE	3723.1	103644.4	0.036	0.971
Weight:ColorF	1757.0	103638.8	0.017	0.986
Weight:ColorG	235.3	103640.5	0.002	0.998
Weight:ColorH	-310.7	103639.0	-0.003	0.998
Weight:ColorI	-833.8	103646.2	-0.008	0.994
Weight:CertifYES	3057.0	103637.2	0.029	0.976
ColorE:CertifYES	7377.6	104163.0	0.071	0.944
ColorF:CertifYES	6354.3	104157.9	0.061	0.951
ColorG:CertifYES	6026.0	104159.2	0.058	0.954
ColorH:CertifYES	5971.8	104158.4	0.057	0.954
ColorI:CertifYES	6023.7	104165.8	0.058	0.954
Weight:ColorE:CertifYES	-8325.9	103647.8	-0.080	0.936
Weight:ColorF:CertifYES	-6466.1	103641.5	-0.062	0.950
Weight:ColorG:CertifYES	-6011.3	103643.3	-0.058	0.954
Weight:ColorH:CertifYES	-5816.6	103642.0	-0.056	0.955
Weight:ColorI:CertifYES	-5666.8	103649.3	-0.055	0.956

```
[...]
```

Interactions: interpretations and recommendations

Three and higher-way interactions should be used with care,

- ▶ They are very difficult to interpret
- ▶ The number of parameters in the model rapidly explodes with the number of variables (and with the number of levels).

Remember that the interaction between factor x_1 and x_2 is the modification of the effect of x_1 (on y) when x_2 is varying (and vice-versa).

In some contexts, interaction is also called moderation because " x_2 moderates the effect of x_1 ".

Interactions: using several models

If interaction between numerical and categorical variables means having several slopes, can't we use several separate regression?

- ▶ It is always possible to make several regressions (subgroup analysis)
- ▶ It is however not completely equivalent to one regression with a categorical variable and interaction.

The difference is that

- ▶ One overall model with interaction makes the assumption that the residuals e have the same structure: $e \sim N(0, \sigma^2)$, in all level of the categorical variable.
- ▶ With several models, each model can have a different structure of the residuals.

Interactions: using several models

When should we use separate regression and when one overall regression? Technical arguments:

- ▶ If standard deviation look inhomogeneous then several regressions is be justified.
- ▶ If standard deviation are homogeneous then a overall model (with the categorical factor!!) is better.

More later in the course about standard deviation analysis. In practice:

- ▶ If the difference between the grouping factor levels is of interest, then one has to use a single regression (with the categorical factor!!).
- ▶ If each subgroup is to be analyzed individually, then separate regression may be simpler.

Interactions: using several models

Examples:

- ▶ In a marketing study, behavior of women and men are both of interest in order to define a strategy of communication. Both are of interest and their difference is not of primary interest: the researchers use two separate regression models.
- ▶ In the study the equality in salary between women and men in a company is of interest. The difference between the two sexes is the central point. An overall model is needed.

Statistical modeling

Module 04

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Autumn 2018

Regression: statistical assumptions

Remember the model assumptions:

- ▶ $y_i = b_0 + b_1x_{1i} + \cdots + b_px_{pi} + e_i$
- ▶ Model errors e_i are statistically independent,
- ▶ The distribution of errors e_i is $N(0, \sigma^2)$

We write

$$e_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

"i.i.d." stands for "independent and identically distributed".

Regression: statistical assumptions

In particular,

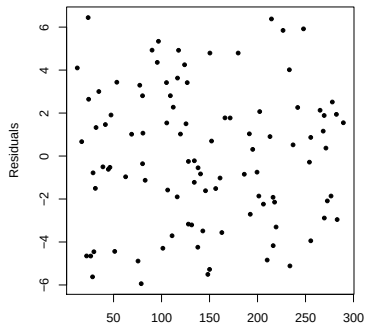
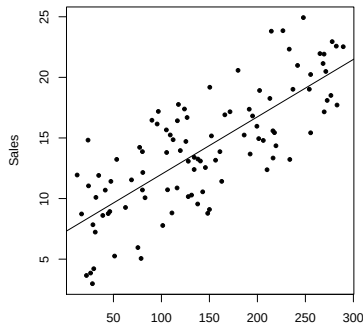
$$E(e_i) = 0, \quad \text{Var}(e_i) = \sigma^2, \quad \text{Cov}(e_i, e_{i'}) = 0, \quad i \neq i'.$$

Consequently,

1. Responses y_i are statistically independent,
2. $y_i \sim N(b_0 + b_1x_{1i} + \cdots + b_px_{pi}, \sigma^2)$.

Regression: statistical assumptions

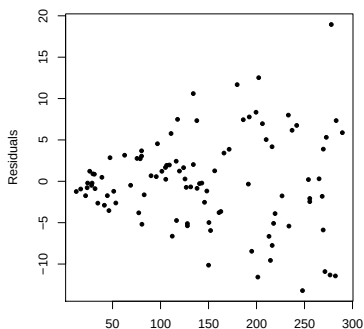
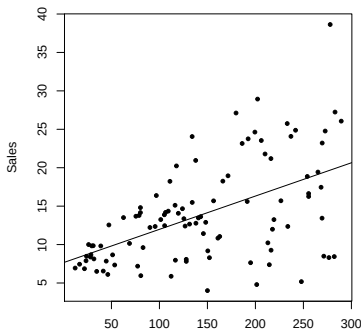
Example: in a company, product sales and TV budgets (local adv. campaign) are studied.



The residuals are normally distributed, or equivalently the observations around the curve.

Regression: statistical assumptions

Example: In that case, the residuals are normally distributed but not with a constant standard deviation. The linear model assumptions are not satisfied.



More later about how to check if these assumptions are satisfied.

The standard deviation σ

We just saw a new very important element of the regression: the standard deviation σ .

Since it is the standard deviation of the model residual, it is

- ▶ A measure of the uncertainty that we have on the response y given x (error on the prediction);
- ▶ A measure of the variability of y that is not explained by the variables in x (residual variance).

A large σ induces both:

- ▶ A large uncertainty of the prediction of y
- ▶ A large uncertainty on the estimation of β .

Its estimation of thus of central importance.

Unbiased estimation of σ^2 : s^2

Consider the sum of squares for the error:

$$SS_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It can be shown that, under the statistical assumptions,

$$E(SS_e) = (n - p - 1)\sigma^2.$$

Thus,

$$s^2 = \frac{1}{n - p - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

is an unbiased estimate of σ^2 :

$$E(s^2) = \sigma^2.$$

Note: $SS_e/(n - p - 1)$ is also called MS_e , Mean Squares (see later).

Unbiased estimation of σ^2 : s^2

Example: Real estate data predicting pricing: $n = 319$ (# products in the database)

$$SS_e = 2.34e^5, MS_e = s^2 = 2.34e^5/313 = 7.48e^2, s = 27.3.$$

Example: With the TV-budget data, (first case) we have

```
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.265872	0.656322	11.07	<2e-16 ***
TVbudg	0.047384	0.003975	11.92	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.177 on 98 degrees of freedom
Multiple R-squared: 0.5919, Adjusted R-squared: 0.5877
F-statistic: 142.1 on 1 and 98 DF, p-value: < 2.2e-16

That is, $s = 3.177$.

Prediction interval

In a situation where we want to predict y given x , instead of getting a point prediction (i.e. a single number) it is much more useful to get an interval of possible values with a certain level of probability.

Under the model assumptions, for a new x , the corresponding unknown y is

$$y \sim N(b_0 + x_1 b_1 + \cdots + c_p b_p, \sigma^2).$$

Thus, for any fixed level α ,

$$P\left\{z_{\alpha/2} \leq \frac{y - (b_0 + x_1 b_1 + \cdots + c_p b_p)}{\sigma} \leq z_{1-\alpha/2}\right\} = 1 - \alpha,$$

Prediction interval

Hence, the prediction interval at level $1 - \alpha$ of y at x is

$$b_0 + x_1 b_1 + \cdots + c_p b_p + z_{\alpha/2} \sigma \leq y \leq b_0 + x_1 b_1 + \cdots + c_p b_p + z_{1-\alpha/2} \sigma.$$

For a more compact formula, we can use the matrix notation:

$$x^T b + z_{\alpha/2} \sigma \leq y \leq x^T b + z_{1-\alpha/2} \sigma,$$

with $x = (1, x_1, \dots, x_p)^T$ and $b = (b_0, b_1, \dots, b_p)^T$.

Estimation of the prediction interval

The previous formula uses the population coefficient b and σ . In practice, replace these unknown quantities by their estimates (plug-in method):

$$x^T \hat{b} + z_{\alpha/2} s \leq y \leq x^T \hat{b} + z_{1-\alpha/2} s.$$

The most used level α is 5%, which provides a prediction interval at 95%.

Estimation of the prediction interval

Example: with the real estate case, for the following product (i.e. a new x),

$$x_1 = 2100 \text{ ft}^2$$

$$x_2 = 15 \text{ yrs}$$

$$x_3 = 4 \text{ bedrooms}$$

$$x_4 = 3 \text{ bathrooms}$$

$$x_5 = 2 \text{ cars in the garage}$$

The predicted price is $\hat{y} = 179$ ($\times 1000\$$).

The estimated prediction interval at 95% is $[125.5, 232.5]$.

Interpretation: in the population, among all the products with characteristics x , 95% have a price in this interval.

Limitations

A prediction interval does not take the estimation error into account. Indeed,

- ▶ $\hat{b}$ is not exactly equal to the true population coefficient b ;
- ▶ s^2 is not exactly equal to the true population variance σ^2 .

Reminder: common values for α and z are

α	$z_{1-\alpha/2}$
1%	2.58
5%	1.96
10%	1.64
20%	1.28

$$z_{\alpha/2} = -z_{1-\alpha/2}$$

Model building: prediction in practice

Given a data base, how can we build a good model? Roughly speaking, two steps:

1. Which independent variables x ?

- ▶ Does it make sense? (e.g. use the price of the butter in Bangladesh to predict unemployment in Switzerland)
- ▶ Is it available? (e.g. not in the data base, very costly)
- ▶ Is it reliable? (e.g. bad data quality, lots of missing values)
- ▶ Is it redundant? (e.g. use weight in kg and in lb)

2. How to include/exclude the variables?

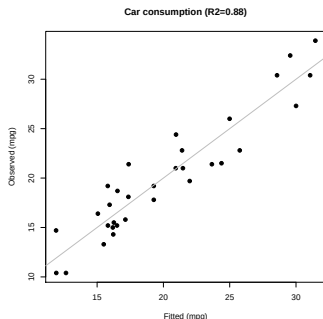
- ▶ Include the variables as they are.
- ▶ Make a change of variable (create a new one).
- ▶ Combine two variables.

Building new variables

One way to build a new variable is to make a change of variable of another one.

Example: with car data, use x_1 is weight, $x_2 = x_1^2$, x_3 is displacement, $x_4 = \log(x_3)$

$$y = 97.8 - 0.97x_1 - 0.21x_2 + 0.048x_3 - 15.8x_4 + e$$

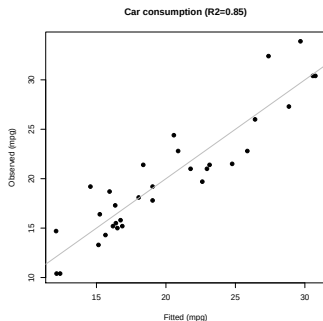


Building new variables

Another possibility is to combine two (or more) existing variables.

Example: x_1 is weight, x_2 is displacement, $x_3 = x_1x_2$,

$$y = 44.1 - 6.5x_1 - 0.06x_2 + 0.12x_3 + e$$



Building new variables

Sorm remarks:

- ▶ As long as parameters b_j are separated by a "+" or "-", it is a linear model. E.g.

$$y = 1 + x_1 + 2x_2 + 5x_2^2 + 3x_1x_2 + 8\log(x_2) + e.$$

- ▶ Linear combinations of variables (e.g. $x_4 = x_2 + 3x_3$) are redundant. E.g.

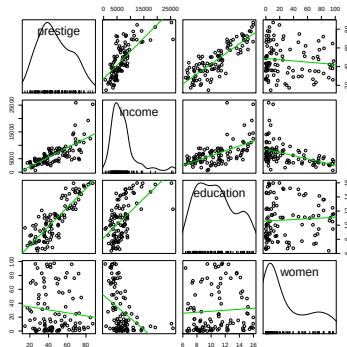
$$\begin{aligned} y &= b_0 + b_1x_1 + b_2x_2 + b_3(x_1 + x_2) + e \\ &= b_0 + (b_1 + b_3)x_1 + (b_2 + b_3)x_2 + e. \end{aligned}$$

Exploratory data analysis: scatter plots

So many models can be built... Use scatter plots to inspect potential links.

Example: find prestige relation to the other variables.

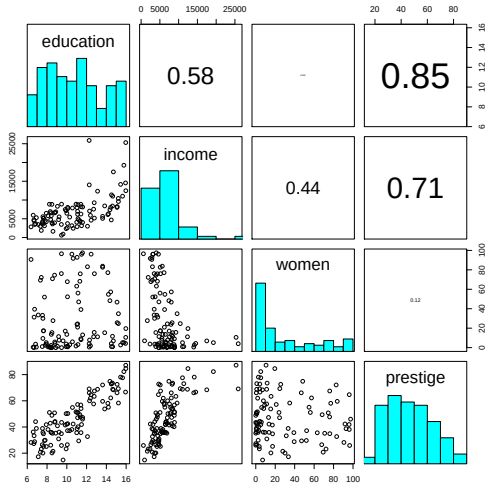
- ▶ education: avg num. of years of edu. for occupational incumbents.
- ▶ income: avg income of occupational incumbents, in \$.
- ▶ women: % of women in the occupation.
- ▶ prestige: avg prestige rating for the occupation.



source: https://rstudio-pubs-static.s3.amazonaws.com/63531_55f83f76f1104efd9b6d694f9228d55e.html

Exploratory data analysis

Use the scatter plot matrix with the correlations



Exploratory data analysis: caveats

Explanatory analysis using correlation and scatter plots are only giving a partial view on the "good" regression model.

- ▶ They are unconditional on the other variables.
E.g. in prestige data set, we see that increases with income, and that income increases with education. We cannot see if prestige would increase with prestige for a fixed level of education (or vice-versa).
- ▶ They are limited to low dimensions. E.g. what if we have 100 variables...

They are only tools for experts to handle a meaningful discussion in their domain.

Exploratory data analysis: recommendations

Recommendations:

1. Use prior expertise to define a meaningful data perimeter of the study (e.g. limit the number of variables).
2. Use explanatory analysis to try to discover how the variables are linked. Use this as guideline only.
3. Define a model and fit it. Inspect its quality (graph and R^2).
4. Review the model if needed: test alternative models¹.

In order to be able to select and use regression models, we need to develop/see some further properties.

¹See later.

Statistical modeling

Module 05

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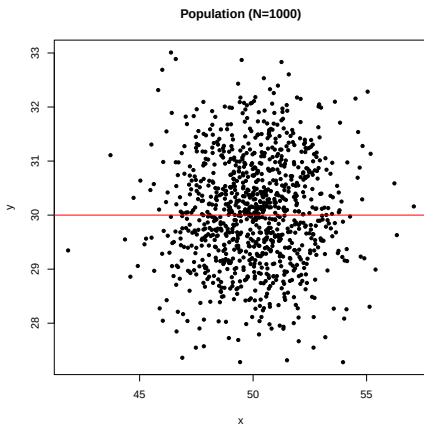
Autumn 2018

Inferential problem

Reminder: inference is about validating/rejecting a hypothesis about a population when we can only observe a sample drawn at random from this population.

Inferential problem

Example: Below the population of times to achieve a task (y) and the volume of the music listen when this task was made (x). These are simulated data. There are independent: y does not vary with x .

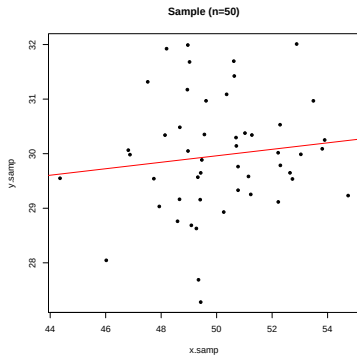


Inferential problem

Suppose that, in a study, the researcher wants to see the effect of the volume x on the time of the task y . The researcher however does not have access to $N = 1000$ observation (the population) but only to a sample of it ($n = 50$), selected at random. The results are:

Model equation:

$$y = 27 + 0.06x + e$$



Inferential problem

From the research point of view (i.e. "is x influencing y , in the population?"), the regression obtained from the sample does not help:

- ▶ The coefficient 0.06 is non-zero, thus x has an impact on y
- ▶ The coefficient is however very small, thus x may not have an real impact.

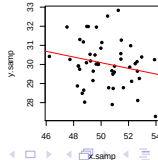
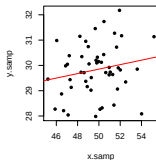
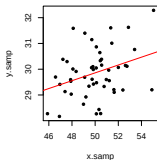
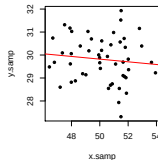
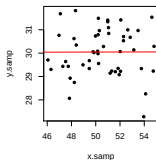
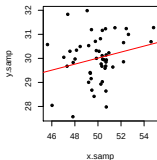
Of course the coefficient cannot be exactly 0. This is just impossible. It is non-null at a given precision level.

In this context, the statistical inference answers the question

"When is a coefficient small enough to conclude that the corresponding variable x has no influence on y ?"

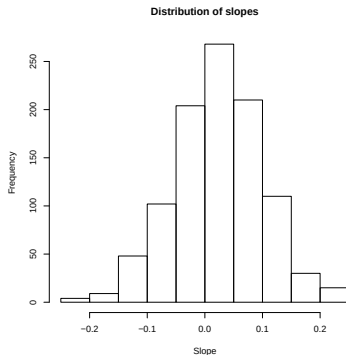
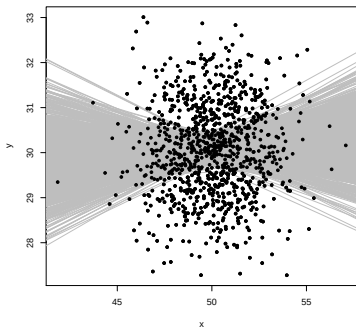
Inferential problem

When the question is related to the population (i.e. that cannot be completely observed), the question should not be "what happened with the sample?" but "what could happen with the sample?". To illustrate, here are 6 possible samples from the population



Inferential problem

If we look at (almost...) all the possible samples that could have been extracted from this population, then we could have



Inferential problem

The distribution of the possible slopes looks like the normal distribution. In fact, knowing that there is no link between x and y in the population, this normality could have been anticipated from the mathematical theory.

Without going into the details of this theory, we will use it to make inference on our regression to draw conclusions not only at the level of the sample but also at the level of the population.

Inferential problem

Mainly two types of problems with regression

- ▶ Prediction: use a sample $\{y_i, x_i\}_{i=1}^n$ already observed to build a trustworthy prediction of y for a new x
- ▶ Research: discover whether a link between y and one or more variables of interest x exists.

If we knew the population model,

$$y = b_0 + b_1x_1 + \cdots + b_px_p + e,$$

then we would easily answer these questions. However, we don't know it. It is only estimated from a sample.

Assuming that the sample is randomly extracted from the population of interest, then one may use statistical theory to answer these questions.

Model significance: F-test

Suppose that, in the population, there is no link at all between y and x . The true population model would thus be $y = b_0 + e$. This is translated into the statistical null hypothesis:

$$H_0 : b_1 = \dots = b_p = 0$$

As illustrated before, using the sample will NEVER give exactly $\hat{b}_1 = \dots = \hat{b}_p = 0$ exactly. So

What could happen if H_0 is true?¹

¹Remember, this is the good question

Model significance: F-test

If H_0 is true then, the mathematical theory tells us that the statistic

$$F = \frac{MS_R}{MS_e} = \frac{SS_R/p}{SS_e/(n - p - 1)}$$

is distributed according to the Fisher distribution

$$F_{df_1, df_2}$$

where the degrees of freedom are

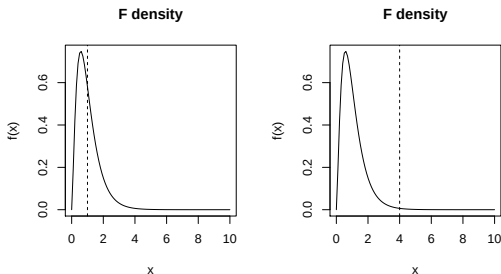
$$df_1 = k, \quad df_2 = n - p - 1.$$

Above, MS stands for "Mean Squares".

Interpretation: if we could repeat the sampling (and the regression fit with the new sample) a very large number of times, computing F for each of these sample, and looking at their distribution, then we would see a Fisher distribution.

Model significance: F-test

Thus, to test if H_0 is supported by the data, the observed value F is compared to the distribution F_{df_1, df_2} , which represents H_0 .



- ▶ Left: the observed F is 1.1. It could come from the F distribution. H_0 and the data are compatibility: do not reject H_0 .
- ▶ Right: the observed F is 4. It is very unlikely F distributed. H_0 and the data are incompatibility: reject H_0 .

Model significance: F-test

For which value of F should we switch from "Do not reject H_0 " to "Reject H_0 "?

The area to the right of F is the p -value

$$p = 1 - F_{df_1, df_2}(F)$$

When this p -value is small (F is large compare to the Fisher distribution) then reject H_0 . The limit of the decision is α , the significance level of the test. Often², $\alpha = 5\%$.

Equivalently, reject H_0 when

- ▶ $p\text{-value} < \alpha$,
- ▶ $F > F_{df_1, df_2; 1-\alpha}$.

²and in this course except mentioned

Model significance: F-test

Procedure

- ▶ Compute $F = \frac{MS_R}{MS_e}$,
- ▶ Compute $F_{df_1, df_2; 1-\alpha}$ (read in table or use R),
- ▶ Reject $H_0 : b_1 = \dots = b_p = 0$ if either

$$F > F_{df_1, df_2; 1-\alpha}$$

or

$$p = 1 - F_{df_1, df_2}(F) < \alpha.$$

In R,

- ▶ $F_{df_1, df_2; 1-a} = \text{qf}(1-a, \text{df1}, \text{df2})$
- ▶ $p = 1 - \text{pf}(F, \text{df1}, \text{df2})$

Model significance: F-test

Example: For the burn out study, restricted to males,

```
> fitM <- lm(BurnOut~Social+I(Social^2), subset=Sex=="M", data=ashley)
> summary(fitM)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	149.59384	187.90193	0.796	0.4521
Social	-4.40025	7.80315	-0.564	0.5904
I(Social^2)	0.15926	0.06968	2.286	0.0562 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 79.47 on 7 degrees of freedom
Multiple R-squared: 0.951, Adjusted R-squared: 0.937
F-statistic: 67.93 on 2 and 7 DF, p-value: 2.604e-05

$df_1 = 2$, $df_2 = 7$, $F = 67.93$. The model is significant at 5% because $F > 4.74 = F_{2,7;0.95}$ (not shown by R). The corresponding p -value is 2.604×10^{-5} .

Model significance: in practice

Whenever a regression model is built, look at the global F -test and its p -value.

- ▶ If H_0 is rejected, use the model as usual.
- ▶ If H_0 is not rejected (at 5%), the model is declared non-significant. The observed link between y and the covariates x is probably due to chance (the sampling).

The F statistics and the R^2 's are intimately linked. Only the F statistic can be used to test the significance of the model.

Even with a small R^2 , a significant model can be used. It is "better than nothing".

ANOVA: ANalysis Of VAriance

The ANOVA table summarizes the F -test calculations. It is often used independently of the regression concept and taken as a theory in itself.

Source	df	SS	MS	F	p
Regression	$df_1 = p$	SS_R	$MS_R = SS_R/df_1$	$F = MS_R/MS_e$	$1 - F_{df_1, df_2}(F)$
Residual	$df_2 = n - p - 1$	SS_e	$MS_e = SS_e/df_2$		
Total	$n - 1$	SS_T			

ANOVA and F-test

Example: Remember the real estate example (pricing of houses). It gives³

Source	df	SS	MS	F	p
Regression	5	$1.04e^9$	$2.1e^8$	277.8	≤ 0.001
Residual	313	$0.23e^9$	$7.5e^5$		
Total	318	$1.27e^9$			

For a test at 5%, use equivalently

- ▶ $277.8 > 3.08 = F_{5,313;0.95}$
- ▶ $p < 0.05$

Therefore reject H_0 , the model is significant (at 5%).

³" e^k " stands for " $\times 10^k$ "

Comparison of two nested models

The F -test is a global test for all the covariates at once. Often a subset of the covariates needs to be tested in order to answer a research question. The F -test can also be used in this case after a small adaptation.

Example: Burn out study: Is there a difference between males and females? To answer, one has to compare models with and without the factor Sex. Consider the two models below:

$$M_1 : y = b_0 + b_1x_1 + b_2x_1^2 + (c_0 + c_1x_1 + c_2x_1^2)\delta_M(x_2) + e$$

versus

$$M_0 : y = b_0 + b_1x_1 + b_2x_1^2 + e$$

where x_1 is the socialization index and x_2 the sex.

Comparison of two nested models

The research hypothesis:

"There is no difference between males and females in the population"

is equivalent to the statistical hypothesis

$$H_0 : c_0 = c_1 = c_2 = 0$$

or also, that the two models are equal: $M_0 = M_1$. Overall, it means that we should not need covariate Sex in the model.

This can be tested using the difference in the sum of squares of each model.

Comparison of two nested models

From the predictions $\hat{y}_{i0}$ and $\hat{y}_{i1}$ of models M_0 and M_1 , the sum of squares of each models are:

$$SS_0 = \sum_{i=1}^n (y_i - \hat{y}_{i0})^2, \quad SS_1 = \sum_{i=1}^n (y_i - \hat{y}_{i1})^2$$

Obviously, $SS_1 \leq SS_0$. Indeed, the full model M_1 is better than M_0 because M_1 has more parameters.

The question is whether this extra complexity (more parameters) is significant.

Comparison of two nested models

Under the model assumptions, it can be shown that, **if H_0 is true**, then the statistics

$$F = \frac{(SS_0 - SS_1)/(df_1 - df_0)}{SS_1/(n - p - 1)} \sim F_{df_1 - df_0, n - p - 1}$$

where

- ▶ n is the sample size,
- ▶ $df_1 = p + 1$ is the number of parameters in model M_1 ,
- ▶ df_0 is the number of parameters in model M_0 .

The test decision is thus to reject H_0 (at level α) if either

- ▶ $F > F_{df_1 - df_0, n - p - 1; 1 - \alpha}$
- ▶ $p\text{-value} = 1 - F_{df_1 - df_0, n - p - 1}(F) < \alpha$

Comparison of two nested models

Example: in the burn out case, the predictions are

$$\begin{aligned}\hat{y}_{i0} &= \hat{b}_{00} + \hat{b}_{01}x_{i1} + \hat{b}_{02}x_{i1}^2 \\ \hat{y}_{i1} &= \hat{b}_{10} + \hat{b}_{11}x_{i1} + \hat{b}_{12}x_{i1}^2 + (\hat{c}_{10} + \hat{c}_{11}x_{i1} + \hat{c}_{12}x_{i1}^2)\delta_M(x_{i2}).\end{aligned}$$

The sum of squares are

$$SS_0 = 361115, SS_1 = 127317, F = 8.57$$

$$n = 20, p = 5, df_1 = p + 1 = 6, df_0 = 3.$$

Thus $F > F_{3,14;0.95} = 3.34$ and H_0 is rejected. Equivalently,

$$p = 1 - F_{3,14}(8.57) = 0.001769 < 0.05.$$

Comparison of two nested models: example

With R,

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Social+I(Social^2), data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

```
Model 1: BurnOut ~ Social + I(Social^2)
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	17	361115				
2	14	127317	3	233797	8.5696	0.001769 **

At 5%, the impact of the socialization index on the Burn Out index is significantly different between Males and Females.

ANOVA in research

Often researchers do not (want to) use regression to perform a test. They are interested into testing the significance of an effect directly. This is why automatic ANOVA procedures have been developed in many statistical computer programs. Lots of confusion can arise from this and we now try to give a clear presentation.

- ▶ When ANOVA is applied to a model, several tests are made.
- ▶ Usually, there are 3 types of ANOVA, sadly called "Sum of type I", "Sum of type II", "Sum of type III".
- ▶ To distinguish which one to use and what is done, the important thing is to understand what is tested (rather than what type of sum is computed).
- ▶ In practice, the most used are type II and type III.

ANOVA: Sum of Type III

The ANOVA of type III is testing if the model with or without the term, keeping all the other terms.

Thus,

- ▶ On row `Social:Sex`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Sex`

- ▶ On row `Social`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Sex + Social:Sex`

- ▶ On row `Sex`, the models are

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Social:Sex`

ANOVA: Sum of Type II

With Anova of type II

```
> Anova(fit, type=2)
Anova Table (Type II tests)

Response: BurnOut
      Sum Sq Df F value    Pr(>F)
Social   874399  1  74.072 2.122e-07 ***
Sex      135742  1  11.499  0.003730 **
Social:Sex 136134  1  11.532  0.003692 **
Residuals 188875 16
[...]
```

Conclusion (test at 5%): keeping all the other terms

- ▶ The interaction is different from 0.
- ▶ That's it...

We could also conclude that, **without the interaction**,

- ▶ The main effect Sex is different from 0,
- ▶ The main effect Social is different from 0.

ANOVA: Sum of Type II

The ANOVA of type II is testing if the model with or without the term, not keeping the interaction terms.

Thus,

- ▶ On row `Social:Sex`, the models are⁵

BurnOut $\sim$ Social + Sex + Social:Sex vs BurnOut $\sim$ Social + Sex

- ▶ On row `Social`, the models are

BurnOut ~ Social + Sex vs BurnOut ~ Sex

- ▶ On row Sex, the models are

BurnOut ~ Social + Sex vs BurnOut ~ Social

⁵same as type III

ANOVA: Sum of Type II or of type III

If interactions are significant, then type II is often misinterpreted.
If interactions are not significant, type III can only be used to infer the interactions, and in general type II should be preferred.

Example:

```
> fit2 <- lm(log(Price)~Certif*Clarity, data=diamond2)
```

```
> Anova(fit2, type=2)
Anova Table (Type II tests)
```

Response: log(Price)

	Sum Sq	Df	F value	Pr(>F)
Certif	24.312	1	49.423	1.412e-11 ***
Clarity	20.755	4	10.548	5.258e-08 ***
Certif:Clarity	3.530	4	1.794	0.13
Residuals	146.590	298		

```
> Anova(fit2, type=3)
Anova Table (Type III tests)
```

Response: log(Price)

	Sum Sq	Df	F value	Pr(>F)
(Intercept)	305.632	1	621.3135	< 2.2e-16 ***
Certif	6.792	1	13.8067	0.0002419 ***
Clarity	0.597	4	0.3035	0.8755167
Certif:Clarity	3.530	4	1.7940	0.1299562
Residuals	146.590	298		

What can be concluded here?

ANOVA: Sum of Type I

The ANOVA of type I is incremental. Read from bottom to top, it test the model with or with all the terms below.

Thus,

- ▶ On row `Social:Sex`, the models are⁶

`BurnOut ~ Social + Sex + Social:Sex` vs `BurnOut ~ Social + Sex`

- ▶ On row `Sex`, the models are⁷

`BurnOut ~ Social + Sex` vs `BurnOut ~ Social`

- ▶ On row `Social`, the models are

`BurnOut ~ Social` vs `BurnOut ~ Sex`

⁶same as type II and III

⁷same as type II

ANOVA: Sum of Type I

```
> anova(fit)
Analysis of Variance Table

Response: BurnOut
          Df Sum Sq Mean Sq F value    Pr(>F)
Social      1  932504   932504   78.994 1.38e-07 ***
Sex         1  135742   135742   11.499 0.003730 **
Social:Sex   1  136134   136134   11.532 0.003692 **
Residuals  16 188875    11805
[...]
```

Conclusion is the same as for type II in this case.

ANOVA of type I is rarely used in practice. Except for balanced design*, it depends on the order of the terms in the model. It is mainly for large models when variable selection is of interest.

It is the result from R function `anova(mod)`. Prefer function `Anova(mod, type 2 or 3)` from package `car`.

ANOVA and F-test: conclusions

In general, prefer a well controlled comparison of models with `anova(mod1, mod2)`.

In practice, declaring not significant a variable can be

- ▶ The purpose of a research study (see later).
- ▶ The occasion to save resources for prediction (e.g. simplify data base in marketing applications, avoid having to track data, etc.).

Statistical modeling

Module 06

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Inference on each coefficient

In the research context, we have seen ANOVA table, F-tests, and model comparisons. These methods allow to test whether a variable is significant in a model.

Research question can also be related to testing some coefficient individually, especially when significance of the difference in levels of categorical variables are of interest.

To do this, we use the t-tests for the coefficients of the regression.

Inference on each coefficient

Example: Whitening power of 4 different toothpastes.

```
> fit <- lm(Whiteness~Toothpaste, data=toothpaste)
> summary(fit)
```

Call:

```
lm(formula = Whiteness ~ Toothpaste, data = toothpaste)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.1667	-1.6250	-0.0833	1.6250	5.8333

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.000	1.419	13.389	1.92e-11 ***
ToothpasteT2	1.667	2.007	0.830	0.416085
ToothpasteT3	9.167	2.007	4.568	0.000187 ***
ToothpasteT4	5.500	2.007	2.741	0.012607 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[...]

Inference on each coefficient

The regression equation is

$$y = b_0 + b_{T2}\delta_{T2}(x) + b_{T3}\delta_{T3}(x) + b_{T4}\delta_{T4}(x) + \epsilon.$$

The estimates can be interpreted as:

- ▶ Average y
 - ▶ for T1 is 19,
 - ▶ for T2 is 20.67,
 - ▶ for T3 is 28.17,
 - ▶ for T4 is 24.5.
- ▶ Average difference between
 - ▶ T2 and T1 is 1.67,
 - ▶ T3 and T1 is 9.17,
 - ▶ T4 and T1 is 5.5.

Inference on each coefficient

The numbers below column $\Pr(>|t|)$ are the p -values of the t-tests of each coefficients. Each one tests if the coefficient is different from 0:

If the p -value is greater than 0.05, the coefficient is declared non-significantly different from zero.

Notes:

- ▶ The significance level 0.05 is ad-hoc. It is the most used value and, except mentioned otherwise, the default in this course.
- ▶ For a quick reading, stars indicate the significance level.
- ▶ A classical language shortcut is to say "the variable is not-significant". But it should be clearly understood that it refers to the coefficient comparison to 0.

Inference on each coefficient

Example: b_{T2} is not significantly different from 0 (at 5%). The others are.

Conclusions: there is no significant difference between the whitening power of T1 and T2. There is a significant difference between T1 and T3, and between T1 and T4.

This kind of conclusions could not have been obtained from an F-test.

On which theory is this test built?

Distribution of the estimates

Suppose that, in the population, the regression is

$$y = b_0 + b_1x_1 + \cdots + b_px_p + \epsilon.$$

In the sample, the (estimated) regression is

$$y = \hat{b}_0 + \hat{b}_1x_1 + \cdots + \hat{b}_px_p + \epsilon.$$

Then it can be shown¹ that ,

$$\hat{b}_j \sim N(b_j, v_{jj}\sigma^2) \quad \text{for } j = 0, \dots, p.$$

where

- ▶ v_{jj} is $(X^T X)^{-1}_{jj}$ the j -th element of the diagonal of $(X^T X)^{-1}$,
- ▶ $v_{jj'}$ is $(X^T X)^{-1}_{jj'}$ the (j, j') -th element of $(X^T X)^{-1}$.

¹Under the model assumptions.

Distribution of the estimates and its estimates...

The estimate of the standard deviation of $\hat{b}_j$, $\sqrt{v_{jj}}\sigma$, is

$$s_{\hat{b}_j} = \sqrt{v_{jj}}s$$

It is called the standard error. It appears below "Std. Error".

The t -value is defined as

$$t_j = \frac{\hat{b}_j}{s\sqrt{v_{jj}}}.$$

It appears below "t value". See the example.

Individual coefficient tests

The null hypothesis of the test H_0 is

$$H_0 : b_j = 0$$

Under H_0 , it can be shown that

$$t_j \sim t_{n-p-1}.$$

A statistical test at level α for

$$H_0 : b_j = 0 \quad \text{versus} \quad H_1 : b_j \neq 0$$

can be built by rejecting H_0 if (equivalent):

- ▶ $|t_j| > t_{n-p-1; 1-\alpha/2},$
- ▶ $p_j = 2 - 2 \times t_{n-p-1}(t_j) < \alpha.$

See column $\Pr(>|t|)$ in the example.

Individual coefficient tests

Example: Another example, the burn out case.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	291.70648	143.20669	2.037	0.0610 .
Social	-4.61486	6.05980	-0.762	0.4590
I(Social^2)	0.10154	0.05730	1.772	0.0981 .
SexM	-142.11265	267.11187	-0.532	0.6030
Social:SexM	0.21461	11.15342	0.019	0.9849
I(Social^2):SexM	0.05772	0.10136	0.569	0.5781

Conclusion: except the slope of Social, at 5%, any coefficient can be set to 0 **keeping all the others.**

Individual coefficient tests

Important remark: the t-test are individuals.

One of the biggest possible mistakes in this course² is to declare **all** the variables not-significant.

The t-test allows to test the significance of each variable **keeping all the others**. In particular, it is possible that among two variables appearing as non-significant with the t-test, at least one would be significant if the other was removed from the model.

For example, it **could be** that `Social` becomes significant if $I(\text{Social}^2)$ was removed from the model. To know this, one needs to try. See later.

²and in the exam...

Individual coefficient tests vs model comparisons

Individual tests cannot be used to test several coefficients at once. For this, use model comparisons or ANOVA instead.

For example, to test if all the coefficients associated to Sex are not significant, compare the models with and without it:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Social+I(Social^2), data=ashley)
> anova(fit0, fit1)
Analysis of Variance Table
```

```
Model 1: BurnOut ~ Social + I(Social^2)
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	17	361115				
2	14	127317	3	233797	8.5696	0.001769 **

Conclusion: we cannot remove **all** the coefficients associated with factor Sex.

Confidence interval

Another important application of the limit distribution is the construction of confidence intervals.

A confidence interval at level $1 - \alpha$ for b_j can be built from the limit distribution of the estimate $\hat{b}_j$:

$$\frac{\hat{b}_j - b_j}{\sigma \sqrt{v_{jj}}} \sim N(0, 1)$$

Therefore, with z_α the α -quantile of $N(0, 1)$,

$$\begin{aligned} 1 - \alpha &= P \left(z_{\alpha/2} \leq \frac{\hat{b}_j - b_j}{\sigma \sqrt{v_{jj}}} \leq z_{1-\alpha/2} \right) \\ &= P \left(\hat{b}_j + z_{\alpha/2} \sigma \sqrt{v_{jj}} \leq b_j \leq \hat{b}_j + z_{1-\alpha/2} \sigma \sqrt{v_{jj}} \right) \end{aligned}$$

Normal confidence interval at $1 - \alpha$:

$$\left[\hat{b}_j + z_{\alpha/2} s \sqrt{v_{jj}}, \hat{b}_j + z_{1-\alpha/2} s \sqrt{v_{jj}} \right].$$

t -confidence interval

Looking carefully at the previous development, the distribution uses σ , while the CI uses its estimate s , since σ is not known.

This has consequences and, in fact, the normal CI is only approximate. A better CI can be built. Indeed, It can be shown that

$$\frac{\hat{b}_j - b_j}{s\sqrt{v_{jj}}} \sim t_{n-p-1}.$$

Thus a confidence interval at $1 - \alpha$ can be built:

$$\left[\hat{b}_j + t_{n-p-1;\alpha/2} s\sqrt{v_{jj}}, \hat{b}_j + t_{n-p-1;1-\alpha/2} s\sqrt{v_{jj}} \right].$$

The confidence interval is interpreted as a range of precision on the estimate of the coefficient with a given confidence level.

t-confidence interval: examples

Toothpaste:

```
> confint(fit)
                2.5 %      97.5 %
(Intercept)  16.039777 21.960223
ToothpasteT2 -2.519721  5.853054
ToothpasteT3  4.980279 13.353054
ToothpasteT4  1.313612  9.686388
```

Burn out:

```
> confint(fit1)
                2.5 %      97.5 %
(Intercept)    -15.4413154 598.8542795
Social          -17.6118401  8.3821200
I(Social^2)      -0.0213515  0.2244359
SexM            -715.0106328 430.7853391
Social:SexM      -23.7070875 24.1363102
I(Social^2):SexM -0.1596726  0.2751168
```

Confidence intervals

Which one to use?

- ▶ The t interval is more accurate.
- ▶ For large n compare to p , the two intervals match quite closely and which one to use is not so important in practice.
- ▶ With a statistical program (R, SPSS, etc.) the t interval is given in general.
- ▶ In practice they often give very close results and the normal one is easier to use (e.g. with a calculator, Excel, etc.).

In any case, the confidence intervals should not be confounded with prediction intervals:

- ▶ Confidence intervals are on the model coefficients (b).
- ▶ Prediction intervals are on the response (y).

Tests of linear combinations of coefficients

Example: For the toothpaste

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	19.000	1.419	13.389	1.92e-11	***
ToothpasteT2	1.667	2.007	0.830	0.416085	
ToothpasteT3	9.167	2.007	4.568	0.000187	***
ToothpasteT4	5.500	2.007	2.741	0.012607	*

One can conclude that T2 is different (at 5%) of T1 (the reference).

How can we compare T2 and T3, for example?

Tests of linear combinations of coefficients

Translate this into statistical hypothesis. First note that:

$$T2 = b_0 + b_{T2}$$

$$T3 = b_0 + b_{T3}$$

Thus, the hypothesis comparing T2 and T3 can be rewritten as

$$T2 = T3 \iff T2 - T3 = 0 \iff b_{T2} - b_{T3} = 0$$

Therefore, we need to be able to test the linear combination $b_{T2} - b_{T3}$ using its estimate $\hat{b}_{T2} - \hat{b}_{T3}$.

Tests of linear combinations of coefficients

Let's look at the distribution of $\hat{b}_{T2} - \hat{b}_{T3}$:

$$\hat{b}_{T2} - \hat{b}_{T3} \sim N(b_{T2} - b_{T3}, \sigma_{T2-T3}^2)$$

where the estimate of σ_{T2-T3}^2 is

$$s_{T2-T3}^2 = s_{\hat{b}_{T2}}^2 + s_{\hat{b}_{T3}}^2 - 2s_{\hat{b}_{T2}}s_{\hat{b}_{T3}}\text{cor}(\hat{b}_{T2}, \hat{b}_{T3}).$$

Under the null hypothesis $H_0 : b_{T2} - b_{T3} = 0$, we have

$$\frac{\hat{b}_{T2} - \hat{b}_{T3}}{s_{T2-T3}} \sim t_{n-p-1}$$

Estimates of the correlations of the estimates...

We see in the previous formula that we need to be able to estimate $cor(\hat{b}_{T2}, \hat{b}_{T3})$, that is the correlation of the estimates.

The covariances and correlations between the coefficient estimates are

$$Cov(\hat{b}_j, \hat{b}_{j'}) = v_{jj'}\sigma^2, \quad Cor(\hat{b}_j, \hat{b}_{j'}) = \frac{v_{jj'}}{\sqrt{v_{jj}v_{j'j'}}}$$

The estimates of the covariances and correlations between the coef. estimates are:

$$\hat{Cov}(\hat{b}_j, \hat{b}_{j'}) = v_{jj'}s^2, \quad \hat{Cor}(\hat{b}_j, \hat{b}_{j'}) = \frac{v_{jj'}}{\sqrt{v_{jj}v_{j'j'}}}$$

Estimates of the correlations of the estimates...

These values appears in the summary of the model in R. Burn out example:

```
> summary(fit1, corr=TRUE)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	291.70648	143.20669	2.037	0.0610 .
Social	-4.61486	6.05980	-0.762	0.4590
I(Social^2)	0.10154	0.05730	1.772	0.0981 .
SexM	-142.11265	267.11187	-0.532	0.6030
Social:SexM	0.21461	11.15342	0.019	0.9849
I(Social^2):SexM	0.05772	0.10136	0.569	0.5781

```
[...]
```

Correlation of Coefficients:

	(Intercept)	Social	I(Social^2)	SexM	Social:SexM
Social	-0.89				
I(Social^2)	0.78	-0.98			
SexM	-0.54	0.47	-0.42		
Social:SexM	0.48	-0.54	0.53	-0.93	
I(Social^2):SexM	-0.44	0.55	-0.57	0.86	-0.98

Tests of linear combinations of coefficients: example

Function `linearHypothesis` allows to extract all these elements and makes the test.

```
> linearHypothesis(fit, hypothesis.matrix="ToothpasteT2 - ToothpasteT3")
Linear hypothesis test
```

Hypothesis:

ToothpasteT2 - ToothpasteT3 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	410.42				
2	20	241.67	1	168.75	13.966	0.0013 **

Conclusion: the two toothpastes T2 and T3 are significantly different at 5%.

Technical details:

- ▶ The above test is a Wald test (somehow equivalent to the *t*-test)
- ▶ Other functions in R exist (`lht`, `glht`, etc.).

Tests of linear combinations of coefficients: general case

To go further...

Any linear combination of the coefficients can be tested:

$$a_0 b_0 + \dots + a_p b_p.$$

Null hypothesis:

$$H_0 : a^T b = 0$$

where $a = (a_0, \dots, a_p)$. The statistic is:

$$\frac{a^T \hat{b}}{s \sqrt{a^T (X^T X)^{-1} a}} \sim t_{n-p-1}$$

Tests of linear combinations of coefficients: general case

Example: another way to test " $T_2 - T_3 = 0$ "

```
> linearHypothesis(fit, hypothesis.matrix=c(0,1,-1,0))
Linear hypothesis test
```

Hypothesis:

ToothpasteT2 - ToothpasteT3 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	410.42				
2	20	241.67	1	168.75	13.966	0.0013 **

Example: Test³ if b_{T_3} is twice as large as b_{T_4} .

```
> linearHypothesis(fit, hypothesis.matrix=c(0,0,1,-2))
Linear hypothesis test
```

Hypothesis:

ToothpasteT3 - 2 ToothpasteT4 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	245.03				
2	20	241.67	1	3.3611	0.2782	0.6037

³Indeed, it is useless; just here for illustration.

Statistical modeling

Module 07

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Regression assumptions

Inference methods (ANOVA, F -tests, coefficient tests, confidence intervals) are all based on some theoretical results. They are valid under probabilistic assumptions about the model.

Departure from these assumptions may have consequences, in particular regarding all the inference:

- ▶ Prediction and confidence intervals could be incorrect,
- ▶ Tests and model comparisons could be not valid.

In order to know if our inferential conclusions are correct, we need to be able to check these assumptions.

Note: the estimation (OLS) is not so dependent on these assumptions. Somehow, a good model can be built with the OLS, even when these assumptions are not validated.

Checking assumptions of the regression

Consider a general model

$$y_i = b_0 + b_1x_{1i} + \cdots + b_px_{pi} + e_i$$

The assumptions about the residual errors e_i are:

- ▶ Mean 0,
- ▶ Variance σ^2 is the same for all e_i (in particular, in all groups, for any value of x , etc.),
- ▶ They are independent, in particular $cov(e_i, e_{i'}) = 0$,

This is overall summarized saying that the residual errors e_i are independent and identically distributed (i.i.d.) according to a normal distribution with mean 0 and variance σ^2 . It is written as

$$e_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

Regression assumptions

The residuals e_i are not directly observed. There are predicted after that the regression is fitted:

$$\hat{e}_i = y_i - \hat{y}_i$$

Under the model assumptions, $\hat{e}_i$ should be close to $N(0, \sigma^2)$, and in particular

- ▶ they should have a mean 0, independent of x .
- ▶ they should have a constant variance, independent of x .
- ▶ they should be normally distributed.

Standardized residuals

However, the $\hat{e}_i$ are not exactly $N(0, \sigma^2)$. Indeed, using the matrix notation

$$\hat{e} = y - \hat{y} = y - X\hat{b} = y - X(X^T X)^{-1}Xy = (I - X(X^T X)^{-1}X)y = Py.$$

P has some properties like $P^2 = P^T P = P$. It can be shown that the so-called standardized residuals, defined as

$$r_i = \frac{\hat{e}_i}{s\sqrt{P_{ii}}},$$

are $N(0, 1)$, if the model assumptions are correct.

Therefore, for the diagnostics, r_i are often preferred to $\hat{e}_i$.

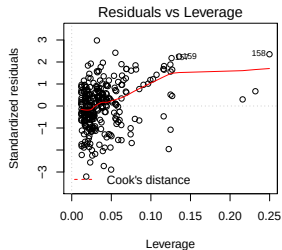
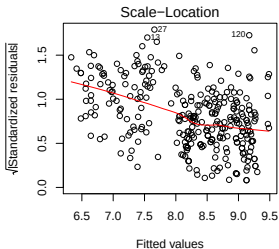
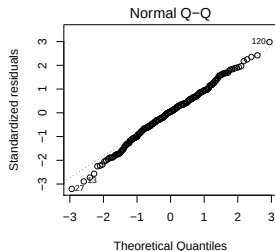
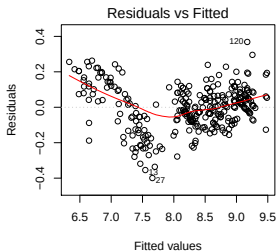
Regression assumptions: 4 graphs in R

To inspect the assumptions on the residuals, several methods exist. A quite popular one is based on specific graphs which helps to make a visual inspection of the residuals to check if the assumptions are satisfied.

It should be noted that these graphs give a general idea. It is not a 0–1 rule.

The default in R is to produce 4 plots.

Regression assumptions: 4 graphs in R



Regression assumptions: 4 plots in R

Top-left: Residuals $\hat{\epsilon}_i$ (y -axis) versus fitted values $\hat{y}_i$ (x -axis).

- ▶ Used to check the "0-mean" assumption:
- ▶ Red line is a visual help.
- ▶ Perfect match with the assumptions: the points should be scattered around the red line, and the red line should be flat.

Regression assumptions: 4 plots in R

Bottom-left: The square root of the absolute value of the standardized residuals $\sqrt{|r_i|}$ (y-axis) versus fitted values $\hat{y}_i$ (x-axis).

- ▶ Used to check the "constant variance" assumption σ^2 .
- ▶ Red line is a visual help.
- ▶ Perfect match with the assumptions: the points should be scattered around the red line, and the red line should be flat.

Regression assumptions: 4 plots in R

Top-right: QQ-plot of the standardized residuals r_i (y -axis) versus the normal distribution quantiles (x -axis).

If $r_i \sim N(0, 1)$, then the proportion of points among $\{r_1, \dots, r_n\}$ that are below z_α should be close to α , for any $0 < \alpha < 1$. This translates into a graph of

- ▶ on the x -axis, $z_{1/(n+1)}, \dots, z_{n/(n+1)}$,
- ▶ on the y -axis, the ordered values $r_{(1)} \leq \dots \leq r_{(n)}$.

These two should match around the $1 - 1$ diagonal.

Regression assumptions: 4 plots in R

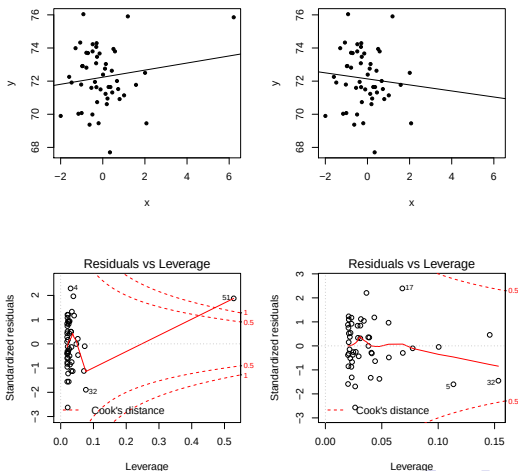
Bottom-right: The standardized residuals r_j versus a leverage index of the corresponding observation (x -axis).

- ▶ Use to detect an influential observation.
- ▶ Influential bounds are in dotted lines (outside the graph range here).
- ▶ An influential data point appears outside these bounds. Such points should be inspected and maybe removed from the data base.

The leverage index explanation is outside the scope of the course.

Regression assumptions: 4 plots in R

The right-most point without being an outlier is influencing a lot the regression line estimation. Its influence is detected by the diagnostic graph. Removing it is a case-by-case issue.



Checking assumptions of the regression: model aptness

Residual analysis is also called model "aptness" analysis.

- ▶ These diagnostics, though technical, are only visual help.
- ▶ They should not be over-interpreted: no black-or-white conclusion.
- ▶ They require a statistical computer program (technical).
- ▶ Often, even when the diagnostics are not completely satisfactory, the model analysis (estimation and inference) remains valid to some extent.
- ▶ These diagnostics are used for general conclusions and also for detecting problematic data points (graph 4).

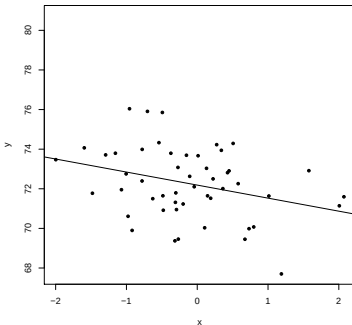
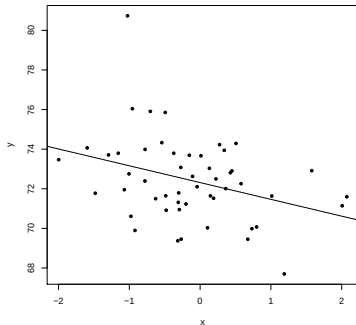
Checking assumptions of the regression: model aptness

In practice,

- ▶ The model is fitted to the data.
- ▶ The analysis of the residuals is performed on the model.
- ▶ If the diagnostics are OK, then the analysis is pursued (tests, ANOVA, etc.).
- ▶ If the diagnostics are very bad, then actions should be taken:
 - ▶ Make Changes of variables (e.g. $\log y$, $\log x$, etc.), or by adding factors to the model (e.g. x^2 , etc.). It is a case-by-case issue.
 - ▶ Some non-parametric alternative exist.
- ▶ If the diagnostics are "average", then the analysis is pursued with some extra caution in particular to p -values that are close to 5%.

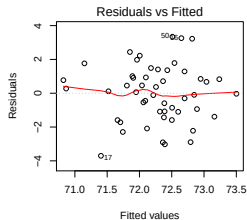
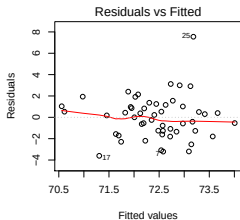
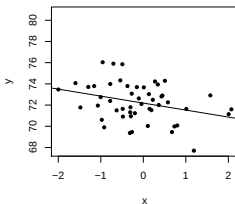
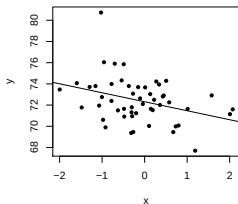
Outliers

The definition of an outlier is quite subjective. It is an observation which, in some ways, is far from what we can expect. Below, observation 25 could be considered as unexpected.



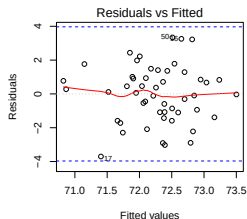
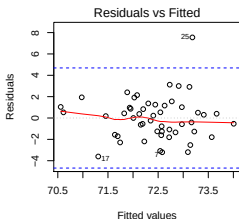
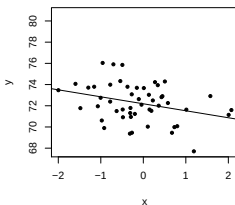
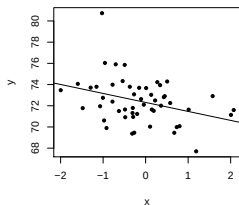
Outliers

One way to detect outliers is to fit a regression line and to inspect the residuals (diagnostic graph 1).



Outliers

In this context, one formal definition of an outlier (with respect to the regression) is an observation such that its residual is outside a range of 99% of the normal distribution.



Outliers

Note that, with this definition, being an outlier is not a "negative" characteristic. It just means that it was unlikely¹ to observed.

The choice of the level 99% was convenient for this example, because there are 51 observations. However, if there are $n = 200$ observations, then we expect to see about 2 outliers.

In this course, we set the level of outliers detection to about $1/(2n)$. Thus, we expect to see less than 1 outlier.

In any case, action to take after an outlier detection is a subjective decision.

¹unlikely according to the model.

Outliers: discussion

- ▶ Regression is a powerful mean to analyze outliers because it can be performed even in large dimension where scatter-plots cannot helps.
- ▶ Influential observation and outliers are not synonymous. An outlier can be very influential. An influential observation need not be an outlier.
- ▶ The presence of influential outliers is problematic since the model may be so badly fitted that the outlier cannot be detected from the residuals.
- ▶ Removing an outlier should not be systematic. In general, its source should be analyzed (error of observation, data contamination, etc.).
- ▶ Robust statistics is the science of developing statistical methods which are robust to the presence of influential observations and outliers.

Statistical modeling

Module 08

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Bachelor, GSEM

Fall 2018

Regression for research

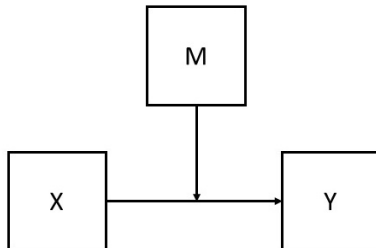
In research, most common questions regarding the study of a response y compare to variables x are:

- ▶ Does a given variable x_j moderates the effect of another variable x_k ? If x_j is large, does the effect on y of x_k different from the one when x_j is small? (Moderation)
- ▶ Does a given factor mediates the effect of another factor? Is the effect of x_j on y direct or just the consequence of its effect on x_k ? (Mediation)
- ▶ Does a given variable x_j has an impact? When x_j varies, does y varies accordingly? (Association and Causality)

Regression for research: Moderation

Moderation is a synonym of interaction. It is often of primary interest in psychology, social science, marketing, etc.

The graphical representation of a moderation effect is



The effect of X on Y is modified when M changes.

Moderation

Example: Burn out case: "Is the effect of socialization on burn out the same for both sexes?"

Compare the models with and without the interactions:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~ Social+I(Social^2) + Sex, data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

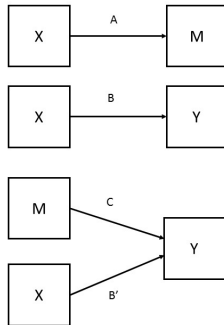
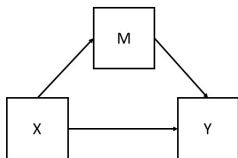
```
Model 1: BurnOut ~ Social + I(Social^2) + Sex
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	16	231845				
2	14	127317	2	104528	5.747	0.01506 *

Answer: yes, factor Sex is a moderator of factor Social (at a significance level of 5%).

Regression for research: Mediation

Mediation means that one factor has its effect partially or fully through the other factor, the mediator. It is a composite hypothesis (i.e. it includes several hypotheses).



Mediation

Mediation means:

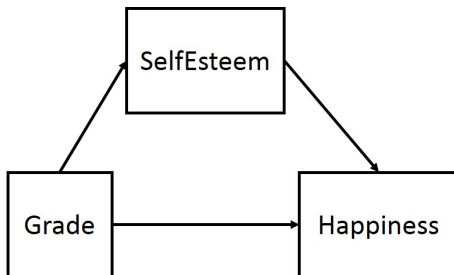
- ▶ Effect A is significant: X has an effect on M .
- ▶ Effect B is significant: X has an effect on Y .
- ▶ Effect B' is lower than B in absolute value: knowing M , the effect of X on Y is smaller (or null).

With mediation, we can conclude that the X has an effect on M which in turn has an effect on Y , or that the effect of X on Y is through M .

Mediation: example

Does self-esteem mediate the effect of grades on happiness?

adapted from <http://data.library.virginia.edu/introduction-to-mediation-analysis/>



Mediation: direct and indirect effects

Indirect and direct effects are often of interest.

If there is a mediation, then the total effect of X on Y is

- ▶ B in the regression $X \rightarrow Y$.
- ▶ $B' + A \times C$ in regression the $X + M \rightarrow Y$.

Both should be equal:

$$B = B' + A \times C.$$

Also

- ▶ $A \times C$ is called the indirect effect of X on Y through M .
- ▶ B' is the direct effect.
- ▶ B is the total effect.

Mediation: example

Three regressions:

```
> summary(fitXY <- lm(Happiness ~ Grades, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.8572	0.6932	4.122	7.88e-05 ***
Grades	0.3961	0.1112	3.564	0.000567 ***

```
[...]
```

```
> summary(fitXM <- lm(SelfEsteem ~ Grades, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.49952	0.58920	2.545	0.0125 *
Grades	0.56102	0.09448	5.938	4.39e-08 ***

```
[...]
```

```
> summary(fitXMY <- lm(Happiness ~ Grades+SelfEsteem, data=myData))
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.9043	0.6055	3.145	0.0022 **
Grades	0.0396	0.1096	0.361	0.7187
SelfEsteem	0.6355	0.1005	6.321	7.92e-09 ***

```
[...]
```

Grades depends on happiness, self-esteem depends on grade, the effect of grades vanishes when self-esteem is considered: it's a full mediation.

Note: to test the significance of the mediation, in fact specific tests should be used.

Mediation: example

In the example,

$$B = 0.3961 = 0.0396 + 0.56102 \times 0.6355 = B' + A \times C$$

- ▶ The total effect is 0.3961
- ▶ The indirect effect is $0.56102 \times 0.6355 = 0.3565$
- ▶ The direct effect is 0.0396.

Considering the non-significance, one could say that the direct effect is 0, and run a regression without Grade

SelfEsteem $\longrightarrow$ Happiness

to compute C and the indirect effect.

Association

Association is not Causality. These two concepts should not be confounded!!

Two variables (or factors) are associated if they are observed varying together.

Correlation (Pearson, Spearman, Kendall) is a measure of association between 2 variables.

Regression helps also describing the association between the response y and a set of variables x . In this context, R^2 's and F -value are measures of association. The coefficients and the p -values are conditional measures of association in the sense that they are estimated given the other variables.

Association and causality

Example: With the burn out data, the researchers want to know if "the socialization index and the burn out index are associated"

From the quadratic model with interaction (`fit1`), the question can be answered by testing the effect of factor `Social` on `BurnOut`. Below with an F -test:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Sex, data=ashley)
> anova(fit0, fit1)
```

Analysis of Variance Table

Model 1: `BurnOut ~ Sex`

Model 2: `BurnOut ~ (Social + I(Social^2)) * Sex`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	18	1199408				
2	14	127317	4	1072090	29.472	1.102e-06 ***

Answer: yes, `BurnOut` and `Social` are associated (at a significance level of 5%).

Association and causality

In previous example, it **cannot** be concluded that factor Socialization **causes** the Burn Out index. We can only conclude that these two factors are **associated**: they have been observed varying together.

Without being too philosophical, causality is a more profound concept that implies a hierarchy into the causes:

x causes y

means that

*a change in x keeping all the other factors constant implies
(causes) a change in y*

Association and causality

Example: in the burn out study, the explanation could be

- ▶ The Burn Out index causes the Socialization: subject with a large burn out probability will little by little de-socialize themselves causing the social index to move.

OR

- ▶ Another unobserved factor causes both Burn Out index and Socialization index. For example, it could be an economical index. A small economical index would imply low job quality, low salary, (etc.) causing both burn out favorable field and de-socialization.

Association and causality

Can we really assess/measure causality? It relies on

- ▶ A design of experiments: observational study are often limited by the randomness of the sampling whereas planned experiments offer a better control on the effect of the factors.
- ▶ A scientific analysis: the answer to the question "what causes what" is often the result of a long process of knowledge acquisition, both in terms of understanding the links, and in terms of which meaningful factors should be included in the study.

In the understanding of causation, statistical analysis is just one tool among others.

Association and causality

Examples: In the burn out study,

- ▶ the fact that burn out causes socialization or the other way cannot be distinguished without a specific analysis and logical reflexion by scientists of the domain. The regression (or any statistical analysis) cannot help here.
- ▶ The fact that there may exist another factor which drives both burn out and socialization indexes cannot be known without an expertise of a scientist in this domain. His/her own knowledge should result from a large number of other studies in this field.

The burn out study is observational. It is a classical limit of observational studies.

Association and causality

Examples (continued): A yield of a production process of some chemical product Y is studied with respect to the quantity of incoming products A and B and the temperature of the cooking chamber. All these factors are perfectly controlled and a plan of experiments is made where several properly chosen combinations of the factors are tested. At the end, it is discovered that the yield increases with the temperature. There, it can be concluded (under the guidance of a chemist) that the temperature causes the yield (in a certain range). Indeed,

- ▶ Since all the factors are perfectly controlled, it is known that their level is not the result of another unobserved factor.
- ▶ Since the temperature is perfectly controlled, it is not the yield which could cause it.

It is an example of a planned study. When they are available, they provide stronger conclusions than observational ones.

Analysis of collinearity

One of the common problems in observational studies is the collinearity of the factors.

Collinearity means that two (or more) factors can vary together so that their effect on a response variable in a regression cannot be distinguished.

This can be detected when there exist correlations (or more generally dependence) between these factors.

Example: The cars example `?mtcars`. The data was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

Analysis of collinearity

A data frame with 32 observations on 11 variables.

- ▶ mpg: Miles/(US) gallon - the response
- ▶ cyl: Number of cylinders (4-6-8)
- ▶ disp: Displacement (cu.in.)
- ▶ hp: Gross horsepower
- ▶ drat: Rear axle ratio
- ▶ wt: Weight (1000 lbs)
- ▶ qsec: 1/4 mile time
- ▶ vs: V-engine/Straight engine (0/1)
- ▶ am: Transmission (0 = automatic, 1 = manual)
- ▶ gear: Number of forward gears
- ▶ carb: Number of carburetors

Analysis of collinearity

```
> summary(lm(mpg~., data=mtcars))
```

Call:

```
lm(formula = mpg ~ ., data = mtcars)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4506	-1.6044	-0.1196	1.2193	4.6271

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.30337	18.71788	0.657	0.5181
cyl	-0.11144	1.04502	-0.107	0.9161
displ	0.01334	0.01786	0.747	0.4635
hp	-0.02148	0.02177	-0.987	0.3350
drat	0.78711	1.63537	0.481	0.6353
wt	-3.71530	1.89441	-1.961	0.0633
qsec	0.82104	0.73084	1.123	0.2739
vs	0.31776	2.10451	0.151	0.8814
am	2.52023	2.05665	1.225	0.2340
gear	0.65541	1.49326	0.439	0.6652
carb	-0.19942	0.82875	-0.241	0.8122

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.65 on 21 degrees of freedom

Multiple R-squared: 0.869, Adjusted R-squared: 0.8066

F-statistic: 13.93 on 10 and 21 DF, p-value: 3.793e-07

Analysis of collinearity

In particular, it means that `cyl` may be predictable with a linear model containing all the other variables (except `mpg`).

If this is true, it means that, approximately,

$$\text{cyl} \approx a_0 + a_1 \text{disp} + \cdots + a_9 \text{carb}$$

If this is true, then `cyl` is useless in the regression on `mpg`

$$\text{mpg} = b_0 + b_1 \text{cyl} + b_2 \text{disp} + \cdots + b_{10} \text{carb} + e$$

which it can be replaced by

$$\text{mpg} = c_0 + c_1 \text{disp} + \cdots + c_9 \text{carb} + e$$

for some $c_0 = b_0 + a_0$, $c_1 = b_2 + c_1 \times a_1$, etc.

In particular, a test on the coefficient b_1 will be non-significant. This is in fact exactly what happened...

Analysis of collinearity

If a research is based on that model (i.e. with collinearities), then the scientific conclusions could be incorrect: `tt` `cyl` and `mpg` are not associated when they are.

"This is so horrible, I can't live with this! How can we avoid that?"

- ▶ Once collinearity is here, this confusion in the conclusion cannot be avoided.
- ▶ Avoiding collinearity can only be done using a design of experiments.
- ▶ At least, we should be informed about collinearity to modulate our conclusions

To analyse collinearity, use the Variance Inflation Factor (VIF).

VIF

The Variance Inflation Factor (VIF) quantifies how much $s_{\hat{b}_j}$, the standard error of the coefficient estimate $\hat{b}_j$, is inflated when multicollinearity exists.

It is computed from the regression for each variable x_j as

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 of the regression of x_j on all the other factors (i.e. x_{-j}).

The interpretation of the VIF is the inflation of the variance of $\hat{b}_j$ due to the collinearity of x_j with the other variables¹. From the previous formulas,

1. If x_j is independent of the other variables then $R_j^2 \approx 0$, and $VIF_j \approx 1$.
2. If x_j is very collinear to the other variables then $R_j^2 \approx 1$, and $VIF_j \approx \infty$.

A rule of thumb commonly admitted is

- ▶ a VIF of 1 means no collinearity.
- ▶ a VIF greater than 4 raises a warning for further investigations.
- ▶ a VIF greater than 10 is a severe sign of collinearity.

¹thus the name, VIF.

Analysis of collinearity: VIF

Example:

```
> fit <- lm(mpg~., data=mtcars)
> vif(fit)
```

cyl	disp	hp	drat	wt	qsec	vs
15.373833	21.620241	9.832037	3.374620	15.164887	7.527958	4.965873
am	gear	carb				
4.648487	5.357452	7.908747				

Thus, collinearity is detected on `cyl`, `disp`, and `wt`. Conclusions from tests on these variables in a research context should be taken with care.

Analysis of collinearity and variable selection

When there is no collinearity, a non-significant variable can be removed from the model. This is called variable selection or model selection. The aim is to build a model which is as simple as possible, as complex as needed.

If there are collinearities, say between x_1 and x_2 , then it should always be kept in mind that removing x_1 from the model is (almost) equivalent to removing x_2 . This is purely arbitrary.

Variable selection: why is it important?

In research,

- ▶ Simplify the model to strengthen the conclusion and provide interpretability (i.e. with lots of variables, conclusion can become very complex to draw).
- ▶ Check whether the sample is representative of a population (e.g. if a variable is known from previous studies but is not selected in the new sample).

In prediction,

- ▶ Get interpretable models,
- ▶ Limit the risk of missing values,
- ▶ Control resources (maintain/track variables can be very costly).

Variable selection

Example: car consumption explained by all the variables.

```
> fit <- lm(mpg ~ ., data=mtcars)
> summary(fit)
```

Call:

```
lm(formula = mpg ~ ., data = mtcars)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4506	-1.6044	-0.1196	1.2193	4.6271

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.30337	18.71788	0.657	0.5181
cyl	-0.11144	1.04502	-0.107	0.9161
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hp	-0.02148	0.02177	-0.987	0.3350
drat	0.78711	1.63537	0.481	0.6353
wt	-3.71530	1.89441	-1.961	0.0633
qsec	0.82104	0.73084	1.123	0.2739
vs	0.31776	2.10451	0.151	0.8814
am	2.52023	2.05665	1.225	0.2340
gear	0.65541	1.49326	0.439	0.6652
carb	-0.19942	0.82875	-0.241	0.8122

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.65 on 21 degrees of freedom

Multiple R-squared: 0.869, Adjusted R-squared: 0.8066

F-statistic: 13.93 on 10 and 21 DF, p-value: 3.793e-07

Variable selection using tests

Individual t-tests show that each variable is not significant **when keeping the others**. However, the model is significant as shown by the F -test. It means that some variables are significant.

Which variables should be kept in the model?

Two approaches:

- ▶ Stepwise selection: remove non-significant variables (at level α) one at a time.
- ▶ Information criterion: select the model with minimum criterion.

Variable selection using tests

In the previous example, for $\alpha = 5\%$, cyl has the highest p -value, we can remove it:

```
> summary(lm(mpg ~ .-cyl, data=mtcars))
[...]  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) 10.96007   13.53030   0.810   0.4266  
disp         0.01283    0.01682   0.763   0.4538  
hp          -0.02191    0.02091  -1.048   0.3062  
drat        0.83520    1.53625   0.544   0.5921  
wt          -3.69251    1.83954  -2.007   0.0572 .  
qsec        0.84244    0.68678   1.227   0.2329  
vs          0.38975    1.94800   0.200   0.8433  
am          2.57743    1.94035   1.328   0.1977  
gear        0.71155    1.36562   0.521   0.6075  
carb       -0.21958    0.78856  -0.278   0.7833
```

Now remove vs, then repeat again and again... Final model:

```
> summary(lm(mpg ~ .-cyl-vs-carb-gear-drat-disp-hp, data=mtcars))
[...]  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)  9.6178     6.9596   1.382 0.177915  
wt           -3.9165     0.7112  -5.507 6.95e-06 ***  
qsec         1.2259     0.2887   4.247 0.000216 ***  
am           2.9358     1.4109   2.081 0.046716 *
```

No more effect can be removed with a test on the coefficients.

Variable selection using tests

The procedure is one of the most used, although it is heuristic.

To be sure, one should do a additional check with a F -test comparing the full model and the simplified one:

```
> fit <- lm(mpg ~ ., data=mtcars)
> fit0 <- lm(mpg ~ .-cyl-vs-carb-gear-drat-disp-hp, data=mtcars)
> anova(fit, fit0)
[...]
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	147.49				
2	28	169.29	-7	-21.791	0.4432	0.8636

The final selection is thus validated: the selected model is (statistically) identical to the most complex one.

Variable selection using tests

Recommendations and comments:

- ▶ Stepwise selection can be used for model growing (adding variables instead of removing them).
- ▶ The rule "remove the highest p-values" is heuristic.
- ▶ At each step any non-significant variable is a candidate for removal, several final models are possible.
- ▶ A scientific prior can also guide into the variable selection: e.g. keep one variable of interest as long as possible.

Selection with tests is an old methods. More and more alternatives are now possible.

Variable selection based on information criterion

Variable selection can be based on an information criterion. The principle is the law of parsimony or Occam's razor:

"Among competing models, the one with the fewest assumptions should be selected"

For statistical models this translates into

"Between models fitting equally well to the data, prefer the simplest one."

A good model should thus achieve a trade-off between goodness-of-fit and simplicity. The AIC measures this tradeoff.

Variable selection based on information criterion

The Akaike Information Criterion (AIC) of a model M is

$$AIC(M) = -2\ell_M + 2K$$

where

- ▶ K is the total number of parameters in model M ,
- ▶ ℓ_M is the maximum log-likelihood of model M .

Variable selection based on information criterion

For regression,

- ▶ $K = p + 2$: intercept, one coefficient per variable (incl. dummy variables), σ .
- ▶ The maximum log-likelihood satisfies

$$-2\ell_M = n \log 2\pi - n \log n + n + n \log SS_e$$

- ▶ The AIC is thus

$$AIC \equiv n \log SS_e - 2(p + 2).$$

It indeed measures a compromise between model complexity ($p + 2$) and model error (SS_e).

The $AIC(M)$ should be as small as possible.

Variable selection based on information criterion

Automatic: At each step, remove the variable for which AIC decreases the most. Stop when no removal decreases AIC anymore.

```
> step(fit, direction="backward")  
Start:  AIC=70.9  
mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb
```

	Df	Sum of Sq	RSS	AIC
- cyl	1	0.0799	147.57	68.915
- vs	1	0.1601	147.66	68.932
- carb	1	0.4067	147.90	68.986
- gear	1	1.3531	148.85	69.190
- drat	1	1.6270	149.12	69.249
- disp	1	3.9167	151.41	69.736
- hp	1	6.8399	154.33	70.348
- qsec	1	8.8641	156.36	70.765
<none>			147.49	70.898
- am	1	10.5467	158.04	71.108
- wt	1	27.0144	174.51	74.280

```
[...]  
Step:  AIC=61.31  
mpg ~ wt + qsec + am
```

	Df	Sum of Sq	RSS	AIC
<none>			169.29	61.307
- am	1	26.178	195.46	63.908
- qsec	1	109.034	278.32	75.217
- wt	1	183.347	352.63	82.790

```
[stop]
```

Variable selection based on information criterion

Other information criteria exist:

- ▶ Bayesian Information Criteria: $BIC = -2\ell_M + K \log n$
- ▶ $AIC_c = AIC + 2K(K + 1)/(n - K - 1)$
- ▶ etc.

Different searches can be performed:

- ▶ Backward: see before;
- ▶ Forward: add one variable at each step;
- ▶ Both: try add or remove at each step.

Variable selection: discussion

When to use which method?

- ▶ In general, stepwise selection with tests is preferred in a research context: the final aim is a test. It is the method which allows a good control on the model selection and gives the highest chance to finish with a meaningful model.
- ▶ In a prediction context (e.g. machine learning, business analytics), AIC (and friends) is preferred. The interpretation of the model is less important, only its prediction capacity and its simplicity are.
- ▶ It is of course possible to remove significant variables with AIC or to have non-significant variable at the end of a AIC selection. These two procedures are just different.
- ▶ Test based or AIC selection can be preceded by a one-by-one variable selection based on the VIF.

Statistical modeling

Module 09

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Bachelor, GSEM

Autumn 2018

Binary outcome

Linear regressions are models for analyzing the links between a set of variables x for a **numerical** outcome y . What can be done when y is not numerical but categorical and more specifically binary?

Example: In a marketing study, one wants to detect if customers are likely to renew their subscription for an online sport information service.

The data set: 60 users, response is $y = 1/0$ (renewed/not renewed), and x 's are

- ▶ Age: age category,
- ▶ AvPages.Week: Avg # of pages viewed per week,
- ▶ Pages.Week: # of pages viewed during the last (previous) week.

Source: [https://help.xlstat.com/customer/en/portal/articles/](https://help.xlstat.com/customer/en/portal/articles/2062235-running-a-logistic-regression-with-xlstat?b_id=9283)

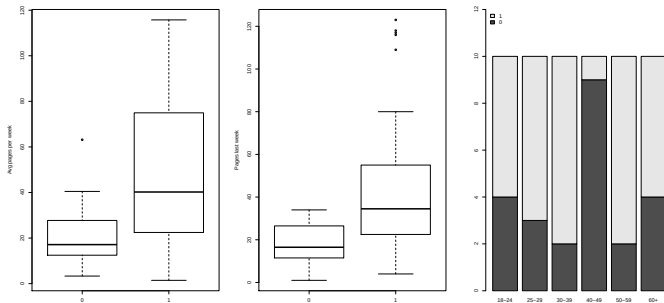
2062235-running-a-logistic-regression-with-xlstat?b_id=9283

Binary outcome

Exploratory analysis:

- ▶ To inspect the relation between y and a numerical x , represent x per level of y ; for example make one boxplot of x for $y = 0$ and one for $y = 1$.
- ▶ To inspect the relation between y and a categorical x , represent a table of frequency of y per level of x ; for example, one bar plot of frequencies of $y = 0/1$ per category of x .

Binary outcome



Renewal ($y = 0/1$) is

- ▶ positively associated with high avg # pages per week (left plot).
- ▶ positively associated with high # pages viewed last week (middle plot).
- ▶ negatively associated with age category 40 – 49 (right plot).

Binary outcome

Exploratory data analysis with a binary outcome has the same limit as with a numerical outcome, namely, that the associations which have been observed are not conditional on the other ones but overall. There may be collinearities here.

Logistic regression is a regression method that allows to estimate/test links between the variables in the same way as regression does for numerical variables.

Logistic regression: concept

Suppose that we can observe a population of kids. We are interested into knowing if they like chocolate or not ($y = 1/0$), according to the sex Girl and Boy $x = B/G$. The result of the analysis would be the proportions per category. For example,

	Boy	Girl
$Y = 1$	$P(Y = 1 B) = 0.65$	$P(Y = 1 G) = 0.53$
$Y = 0$	$P(Y = 0 B) = 0.35$	$P(Y = 0 G) = 0.47$

Of course, in general, we cannot observe the whole population and cannot know these numbers. We need to draw a sample at random.

Logistic regression: concept

Drawing an observation at random (e.g. selecting a kid and asking him whether he/she likes chocolate), the response Y can be 1 or 0.

The probability depends on the variable x (e.g. the sex of the kid). That is, the response Y are distributed according to a Bernoulli distribution with parameter $p(x)$, where x are the independent variables:

$$P(Y = 1|x) = p(x), \quad P(Y = 0|x) = 1 - p(x)$$

In the example,

$$P(Y = 1|B) = p(B) = 0.65, \quad P(Y = 1|G) = p(G) = 0.53.$$

$$P(Y = 0|B) = 1 - p(B) = 0.35, \quad P(Y = 0|G) = 1 - p(G) = 0.47.$$

Logistic regression: linear predictor

With logistic regression, the probability $p(x)$ is linked with x through the linear predictor $\eta(x)$

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p.$$

In particular, the predicted probability given x is

$$p(x) = \frac{\exp\{\eta(x)\}}{1 + \exp\{\eta(x)\}}$$

The link function, transforming $p(x)$ in $\eta(x)$ is called the logit function

$$\eta(x) = \log \left(\frac{p(x)}{1 - p(x)} \right).$$

Logistic regression: example

Example: For the kids example,

$$\eta(B) = \log \left(\frac{0.65}{1 - 0.65} \right) = 0.62.$$

and

$$\eta(G) = \log \left(\frac{0.53}{1 - 0.53} \right) = 0.12.$$

Using a dummy variable notation with G as the reference, we can write

$$\eta(x) = b_0 + b_1 \delta_B(x) = 0.12 + 0.5 \times \delta_B(x).$$

Logistic regression: interpretation

Coefficient $b_1 = 0.5$ quantifies the difference between girls and boys in terms of the probability that $Y = 1$. It plays the role of a contrast like in regression. In particular,

- ▶ if $b_1 = 0$, there would be no difference between boys and girls.
- ▶ if $b_1 > 0$, then $p(B) > p(G)$
- ▶ if $b_1 < 0$, then $p(B) < p(G)$

Logistic regression: prediction

For a new x , the prediction $\hat{y}(x)$ is obtained by predicting the linear predictor, the probability, and then the outcome:

1. The linear predictor

$$\hat{\eta}(x) = \hat{b}_0 + \hat{b}_1 x_1 + \cdots + \hat{b}_p x_p$$

2. The probability

$$\hat{p}(x) = \frac{\exp\{\hat{\eta}(x)\}}{1 + \exp\{\hat{\eta}(x)\}}$$

3. The outcome

$$\hat{y}(x) = \begin{cases} 0, & \text{if } \hat{p}(x) < 0.5 \\ 1, & \text{if } \hat{p}(x) \geq 0.5 \end{cases}$$

Logistic regression: example

Example: with the marketing study example the estimate is

```
> renew.mod <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
  family="binomial")
> summary(renew.mod)
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.35667	1.24579	-1.892	0.0585 .
Pages.Week	0.08928	0.03564	2.505	0.0123 *
AvPages.Week	0.02350	0.02585	0.909	0.3634
Age25-29	0.68781	1.20624	0.570	0.5685
Age30-39	0.96324	1.27573	0.755	0.4502
Age40-49	-2.98274	1.37732	-2.166	0.0303 *
Age50-59	1.08553	1.16754	0.930	0.3525
Age60+	0.30892	1.26437	0.244	0.8070

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

[...]

Logistic regression: prediction

The fitted linear predictor is

$$\begin{aligned}\hat{\eta}(x) = & -2.4 + 0.09x_1 + 0.02x_2 + 0.69\delta_{g_1}(x_3) \\ & + 0.96\delta_{g_2}(x_3) - 2.98\delta_{g_3}(x_3) + 1.09\delta_{g_4}(x_3) \\ & + 0.31\delta_{g_5}(x_3)\end{aligned}$$

where

- ▶ x_1 is "pages last week",
- ▶ x_2 is "Avg pages per week",
- ▶ x_3 is "Age" with levels $g_0 = "18 - 24"$ (the reference), $g_1 = "25 - 29"$, $\dots$, $g_5 = "60 + "$.

Logistic regression: prediction

For a customer with characteristics $x_1 = 30$, $x_2 = 25$, $x_3 = "18 - 24"$, the prediction is thus

$$\hat{\eta}(x) = -2.4 + 0.09 \times 30 + 0.02 \times 25 = 0.91,$$

$$\hat{p}(x) = e^{0.91} / (1 + e^{0.91}) = 0.71$$

$$\hat{y}(x) = 1$$

If the customer is in age category $x_3 = "40 - 49"$, then

$$\hat{\eta}(x) = -2.4 + 0.09 \times 30 + 0.02 \times 25 - 2.98 = -2.07,$$

$$\hat{p}(x) = e^{-2.07} / (1 + e^{-2.07}) = 0.11$$

$$\hat{y}(x) = 0$$

These examples show us how the probability varies when switching level for the categorical variable.

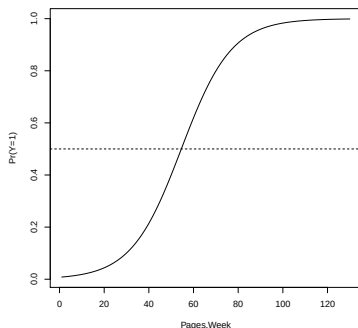
Logistic regression: interpretation

In the above example, the model has estimated that

- ▶ Within the population of customers with $x_1 = 30$, $x_2 = 25$, $x_3 = "18 - 24"$, the proportion of renewing customers is 71% (and 29% not renewing),
- ▶ Within the population of customers with $x_1 = 30$, $x_2 = 25$, $x_3 = "40 - 49"$, the proportion of renewing customers is 11% (and 89% not renewing).

Logistic regression: interpretation

Furthermore, let's consider the sub-population of customers with $x_2 = 20$, $x_3 = "40 - 49"$. The proportions of renewing customers ($y = 1$) in function of x_1 is estimated by the model as



The dashed line at 0.5 is the limit at which the prediction switch from $\hat{y}(x) = 0$ (below) to $\hat{y}(x) = 1$ (above).

Logistic regression: interaction

Interaction effects can be included in the model and are interpreted exactly the same way as in regression: "how the level of one variable moderates the effect of another one"

Example: with the previous model, the effect of `Pages.Week` was estimated to 0.08928. The researchers suspect that this effect varies according to the age. They include an interaction term between `Age` and `Pages.Week` in the logistic regression.

Logistic regression: interaction

```
> renew.mod2 <- glm(Renewed~Pages.Week*Age + AvPages.Week, data=Renew,  
                    family="binomial")
```

```
> summary(renew.mod2)
```

```
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.475e-01	1.337e+00	-0.484	0.628
Pages.Week	2.470e-02	4.430e-02	0.557	0.577
Age25-29	-1.530e+00	2.218e+00	-0.690	0.490
Age30-39	-6.538e-01	2.041e+00	-0.320	0.749
Age40-49	-1.334e+02	1.894e+04	-0.007	0.994
Age50-59	2.277e-01	2.434e+00	0.094	0.925
Age60+	-2.022e+00	2.187e+00	-0.925	0.355
AvPages.Week	1.486e-02	2.517e-02	0.590	0.555
Pages.Week:Age25-29	1.223e-01	1.221e-01	1.001	0.317
Pages.Week:Age30-39	7.373e-02	8.744e-02	0.843	0.399
Pages.Week:Age40-49	3.412e+00	5.038e+02	0.007	0.995
Pages.Week:Age50-59	3.012e-02	1.005e-01	0.300	0.764
Pages.Week:Age60+	1.323e-01	1.441e-01	0.918	0.359

```
[...]
```


Logistic regression: interaction

Because Age has lots of levels, the model has now a lot of coefficients. The interpretation remains the same as for a regression. Considering the slope in front of variable `Pages.Week` (x_1), then

- ▶ If Age is "18-24" then it is $0.02470x_1$
- ▶ If Age is "25-29" then it is $(0.02470 - 0.1223)x_1$
- ▶ If Age is "30-39" then it is $(0.0246 - 0.07373)x_1$
- ▶ etc.
- ▶ If Age is "60+" then it is $(0.0246 - 0.1323)x_1$.

Question: looking at the coefficients, in which age category is the slope of x_1 the highest? The lowest?

Logistic regression: prediction "interval"

For logistic regression, there is no equivalent to the prediction interval of the regression. It is rather obtained by giving the estimation of the probability $\hat{p}(x)$.

Examples: Suppose that the predictions of a customer are

1. $\hat{y} = 1$ with $\hat{p} = 0.99$: the prediction is quite certain.
2. $\hat{y} = 1$ with $\hat{p} = 0.53$: the prediction is quite uncertain.

In the case of regression, with the same prediction $\hat{y}$,

- ▶ Case $\hat{p} = 0.99$ would correspond to a narrow prediction interval.
- ▶ Case $\hat{p} = 0.53$ would correspond to a wide prediction interval.

Logistic regression: estimation

For a regression, the estimate $\hat{b}$ is the OLS.

For logistic regression, it is the maximum likelihood estimator (MLE).

This principle is to get $\hat{b}$ such that the model gives the maximum probability of observing what was actually observed. Consider the simple example below.

Logistic regression: estimation

With $n = 2$ (two observations), assume that the observed sample is $y_1 = 0$, $y_2 = 1$ with x_1 and x_2 given. Under the model (Bernoulli), the probability of observing this sample is

$$\begin{aligned} P(\text{"observe this sample"}) &= P(y_1 = 0, y_2 = 1) \\ &= P(y_1 = 0)P(y_2 = 1) \\ &= \{1 - p(x_1)\}p(x_2) \\ &= \frac{1}{1 + \exp\{\eta(x_1)\}} \frac{\exp\{\eta(x_2)\}}{1 + \exp\{\eta(x_2)\}} \end{aligned}$$

Logistic regression: MLE

The maximum likelihood principle says that one should select $\hat{b}$ which maximizes this quantity, which is called the likelihood.

That is

$$\hat{b} = \arg \max_b L(b),$$

where

$$L(b) = \frac{\exp\{\eta(x_2)\}}{(1 + \exp\{\eta(x_1)\})(1 + \exp\{\eta(x_2)\})},$$

and

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p.$$

The estimator $\hat{b}$ hence obtained is called the maximum likelihood estimator (MLE).

Logistic regression: MLE

In practice, to obtain the MLE, it is easier and equivalent to maximize the logarithm of the likelihood, so called the log-likelihood:

$$\hat{b} = \arg \max_b \ell(b),$$

where

$$\ell(b) = \eta(x_2) - \log \left\{ 1 + e^{\eta(x_1)} \right\} - \log \left\{ 1 + e^{\eta(x_2)} \right\}$$

Note: The maximum likelihood is a general principle that can be applied to lots of model. E.g. when applied to linear regression, the OLS is obtained.

Logistic regression: MLE

In general, for a sample of size n :

$$\hat{b} = \arg \max_b \sum_{i=1}^n y_i \eta(x_i) - \sum_{i=1}^n \log \left\{ 1 + e^{\eta(x_i)} \right\}$$

where

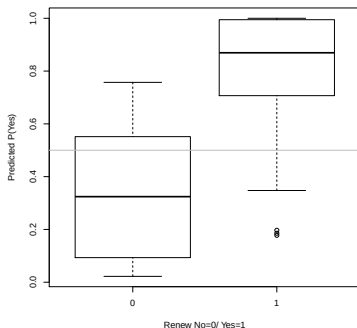
$$\eta(x_i) = b_0 + b_1 x_{i1} + \cdots + b_p x_{ip}.$$

To solve this, one uses numerical algorithms like Fisher scoring and Iterative Re-weighted Least Squares. The practical consequence is that one needs mathematical or statistical computer programs to use logistic regression.

Logistic regression: inspection of the prediction quality

For linear regression, the graphical way to inspect the quality of the prediction is the plot of the observed value y against the fitted values $\hat{y}$.

For logistic regression this is replaced by two boxplots of $\hat{p}(x_i)$, one for $y_i = 0$ and one for $y_i = 1$.



Logistic regression: inspection of the prediction quality

- ▶ On the y -axis, the fitted probability, on the x -axis, the observed class.
- ▶ The misclassified observations are above 0.5 for $y = 0$ and below 0.5 for $y = 1$.

The more the two boxplots are separated away from 0.5, the better the model.

Indeed, if the boxplots are very well separated, then it means that the fitted probabilities are either large (close to 1) or low (close to 0) and thus the model predicts the class precisely.

Logistic regression: inspection of the prediction quality

Unlike for linear regression, there is no R^2 's statistics for logistic regression.

A numerical way of inspecting the model quality is to build the so-called confusion matrix. Below

- ▶ TP stands for "True Positive" and FP for "False Positive",
- ▶ TN stands for "True Negative" and FN for "False Negative".

	Pred. 1	Pred. 0
Obs.1	TP	FN
Obs. 0	FP	TN

Logistic regression: inspection of the prediction quality

Example: with marketing data, "1" means "Renewed".

	Pred. 1	Pred. 0	Total
Obs.1	30 (TP)	6 (FN)	36
Obs. 0	7 (FP)	17 (TN)	24
Total	37	23	60

The accuracy is the proportion of correct prediction. It's a score for the model quality:

$$A = \frac{30 + 17}{60} = 78.3\%$$

Logistic regression: inspection of the prediction quality

Accuracy is a rough measure. The two following conditional probabilities are often inspected:

- ▶ Sensitivity: probability of predicting 1 given it is indeed 1

$$\frac{TP}{TP + FN} = P(\text{"Pred. 1"} | \text{"Obs. 1"}) = \frac{30}{36}$$

- ▶ Specificity: probability of predicting 0 given it is indeed 0

$$\frac{TN}{TN + FP} = P(\text{"Pred. 0"} | \text{"Obs. 0"}) = \frac{17}{24}$$

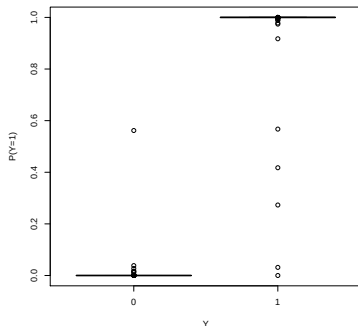
A good model has a large sensitivity and a large specificity¹.

¹Please don't ask how much is large...

Logistic regression: ROC curve

A finer inspection tool of the logistic regression quality is the ROC curve (Receiver Operating Characteristic). The idea is the following.

Consider a very good model like below.



The sensitivity and specificity will be good for sure.

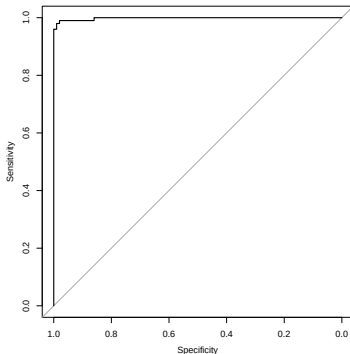
Logistic regression: ROC curve

In fact, these will also be very good, even with a threshold of prediction different from 0.5. In fact, for almost any threshold (btw 0 and 1), the sensitivity and the specificity will be good. This is a sign that the two categories 0 and 1 are very well separated by the model.

The ROC curve represent the sensitivity (y -axis) versus the specificity (x -axis; inverted).

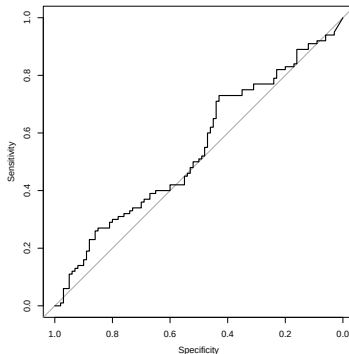
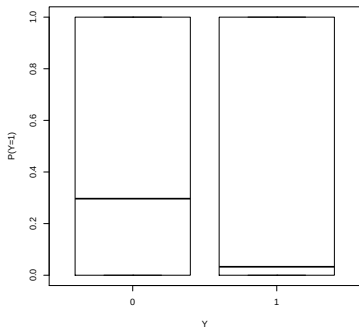
Logistic regression: ROC curve

A good model achieves both large sensitivity and specificity at all threshold levels (like below).



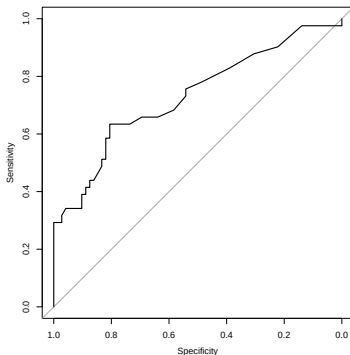
Logistic regression: ROC curve

A bad model (random prediction) achieves a ROC curve close to the diagonal.



Logistic regression: ROC curve

An average model achieves a ROC curve in between the upper corner and the diagonal.



Logistic regression: AUC

AUC stands for "Area Under the Curve". Since the ROC curve identifies good model when they are closer to the left upper corner, one can also compute the area under the ROC curve. The closest to 1 the better the model. An AUC of 0.5 is bad (ROC is diagonal).

For the 3 previous examples:

- ▶ Good: 0.998
- ▶ Bad: 0.538
- ▶ Avg: 0.731

Statistical modeling

Module 10

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Bachelor, GSEM

Autumn 2018

Logistic regression: Comparisons of models

Like with the F -test for regressions, logistic regression models can be compared using a statistical test.

However, there is no sum of squares because they are meaningless for binary (0/1) outcome. The sum of squares can be replaced by the log-likelihood value and the F -test by the so-called likelihood ratio test.

Remark: Except from the fact that the elements in the calculations are different because the underlying theory is slightly different, this method is used **exactly** in the same way as for regression. In particular, p -values are produced and models can be compared.

Logistic regression: Comparisons of models

Suppose we want to compare two models M_1 and M_0 . M_1 is the most complex model (it has more variables) and M_0 is nested within M_1 (i.e. it has the same variables except some of them). Then we have the following notations

- ▶ Models M_1 and M_0 , M_0 nested within M_1 ,
- ▶ Number of parameters in each models: $p_1 > p_0$,

To estimate the two models, the MLE is used, and the corresponding maximum log-likelihood are computed:

$$\ell_0(\hat{b}_0) = \arg \max_{M_0} \ell(b), \quad \ell_1(\hat{b}_1) = \arg \max_{M_1} \ell(b)$$

Logistic regression: Comparisons of models

Since M_0 is nested within M_1 , its maximum log-likelihood is smaller than the one of M_0 ,

$$\ell_0(\hat{b}_0) \leq \ell_1(\hat{b}_1).$$

The analogy for regression is that the sum of square of M_1 is always smaller than the sum of squares of M_0 .

As a consequence, the following difference is always non-negative

$$2 \left\{ \ell_1(\hat{b}_1) - \ell_0(\hat{b}_0) \right\} \geq 0.$$

The log-likelihood playing the same role as the sum of squares, the limit of this inequality,

$$\ell_1(\hat{b}_1) = \ell_0(\hat{b}_0),$$

corresponds to the two models are the same $M_0 = M_1$.

Logistic regression: Comparisons of models

In order to test the null hypothesis $H_0 : M_0 = M_1$, the following test plays the same role as the F -test for the regression.

The Likelihood Ratio Test (LRT) states that, if $H_0 : M_0 = M_1$ is true, then the following difference follows a $\chi^2_{p_1-p_0}$ distribution:

$$LRT = 2 \left\{ \ell_1(\hat{b}_1) - \ell_0(\hat{b}_0) \right\} \sim \chi^2_{p_1-p_0}$$

The test decision is thus to reject H_0 (at level α) if either

- ▶ $LRT > \chi^2_{p_1-p_0; 1-\alpha}$,
- ▶ $p\text{-value} = 1 - \chi^2_{p_1-p_0}(LRT) < \alpha$.

Logistic regression: Comparisons of models

Example: Is factor Age significantly (5%) associated with the response Renewed?

Like for the regression, the test can be done using function `anova`:

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
              family="binomial")
> mod0 <- glm(Renewed~Pages.Week + AvPages.Week, data=Renew,
              family="binomial")
> anova(mod0,mod1, test="LRT")
```

Analysis of Deviance Table

Model 1: Renewed ~ Pages.Week + AvPages.Week

Model 2: Renewed ~ Pages.Week + AvPages.Week + Age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	57	61.203			
2	52	46.943	5	14.261	0.01404 *

Test: since $0.014 < 0.05$, H_0 is rejected. Thus Age is associated with Renewed.

Logistic regression: Comparisons of models

In details, R is computing the following elements

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
               family="binomial")
> mod0 <- glm(Renewed~Pages.Week + AvPages.Week, data=Renew,
               family="binomial")
> (ell0 <- logLik(mod0))
'log Lik.' -30.60175 (df=3)
> (ell1 <- logLik(mod1))
'log Lik.' -23.47138 (df=8)
> (LRT <- 2*(ell1 - ell0))
'log Lik.' 14.26075 (df=8)
> qchisq(0.95, df=5)
[1] 11.0705
```

That is, since $14.3 > 11.1$, H_0 is rejected (i.e. Age is significant at 5%).

Logistic regression: Comparisons of models

In the above example, the two models are

$$M_1 : \eta_1(x) = b_0 + b_1x_1 + b_2x_2 + b_{31}\delta_{g_1}(x_3) + b_{32}\delta_{g_2}(x_3) \\ + b_{33}\delta_{g_3}(x_3) + b_{34}\delta_{g_4}(x_3) + b_{35}\delta_{g_5}(x_3)$$

$$M_0 : \eta_0(x) = b_0 + b_1x_1 + b_2x_2.$$

The null hypothesis $H_0 : M_0 = M_1$ corresponds to

$$H_0 : b_{31} = b_{32} = b_{33} = b_{34} = b_{35}$$

that is,

" H_0 : all the coefficients associated to Age are 0"

or equivalently Age is not significant (i.e. there is no difference between age categories in terms of renewal)

Logistic regression: Comparisons of models

The number of parameters are respectively

- ▶ $p_0 = 3$

- ▶ $p_1 = 8$

And thus, indeed,

$$p_1 - p_0 = 5$$

which corresponds to the number of coefficients associated with Age and that are tested.

Logistic regression: model significance

In the case of a regression, the overall model significance was checked by an F -test. This may be done also for logistic regression by comparing this model (M_1) to the null model (M_0), i.e. containing only the intercept.

Unlike for linear regression, where the model significance is always checked by default at the moment of model fit, this is not automatic for logistic regression. It thus needs to be performed by the user by hand.

Logistic regression: model significance

Example: the full model is significant at 5%

$$M_1 : \eta_1(x) = b_0 + b_1x_1 + b_2x_2 + b_{31}\delta_{g_1}(x_3) + b_{32}\delta_{g_2}(x_3) \\ + b_{33}\delta_{g_3}(x_3) + b_{34}\delta_{g_4}(x_3) + b_{35}\delta_{g_5}(x_3)$$

$$M_0 : \eta_0(x) = b_0.$$

```
> mod1 <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,  
              family="binomial")
```

```
> mod0 <- glm(Renewed~1, data=Renew, family="binomial")
```

```
> anova(mod0,mod1, test="LRT")
```

Analysis of Deviance Table

Model 1: Renewed ~ 1

Model 2: Renewed ~ Pages.Week + AvPages.Week + Age

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	59	80.761			
2	52	46.943	7	33.819	1.862e-05 ***

Logistic regression: individual tests on the coefficients

Continuing the analogy with linear regression, tests on the coefficients can be made. The underlying mathematical results are similar to the one for the regression. It states that, if b_j is the true coefficient value in the population, then

$$\frac{\hat{b}_j - b_j}{s_{\hat{b}_j}} \sim N(0, 1)$$

where

- ▶ $p + 1$ is the number of parameters in the model,
- ▶ " $\sim N(0, 1)$ " states for "is approximately distributed as a $N(0, 1)$ ",
- ▶ $s_{\hat{b}_j}$ is an estimate of the standard error of $\hat{b}_j$, which is a by-product of the fitting algorithm.

Logistic regression: tests on coefficients

Using the previous result, we have a test on one coefficient (**keeping all the other terms**), i.e. we test

$$H_0 : b_j = 0 \quad \text{versus} \quad H_1 : b_j \neq 0.$$

It uses the t -values

$$t_j = \hat{b}_j / s_{\hat{b}_j}, \quad j = 1, \dots, p$$

The test rejects H_0 at level α if either (equivalently)

- ▶ $|t_j| > z_{1-\alpha/2}$
- ▶ $p_j = 2 - 2 \times \text{pnorm}(t_j) < \alpha$

Logistic regression: tests on the coefficients

```
> summary(renew.mod)
Call:
glm(formula = Renewed ~ Pages.Week + AvPages.Week + Age, family = "binomial",
    data = Renew)
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.35667	1.24579	-1.892	0.0585 .
Pages.Week	0.08928	0.03564	2.505	0.0123 *
AvPages.Week	0.02350	0.02585	0.909	0.3634
Age25-29	0.68781	1.20624	0.570	0.5685
Age30-39	0.96324	1.27573	0.755	0.4502
Age40-49	-2.98274	1.37732	-2.166	0.0303 *
Age50-59	1.08553	1.16754	0.930	0.3525
Age60+	0.30892	1.26437	0.244	0.8070

```
[...]
```


Logistic regression: tests on the coefficients

It can be concluded that, at 5%,

- ▶ Factor `Pages.Week` is significant.
- ▶ Factor `AvPages.Week` is not significant **keeping all the other terms**.
- ▶ There is no significant difference between Age levels "18-24" and "25-29" (**keeping all the other terms**).
- ▶ There is no significant difference between Age levels "18-24" and "30-39" (**keeping all the other terms**).
- ▶ etc.

Logistic regression: confidence intervals

The theoretical t -value distribution allows to obtain an approximate confidence interval at $1 - \alpha$:

$$\begin{aligned} 1 - \alpha &= P \left(-z_{1-\alpha/2} \leq \frac{\hat{b}_j - b_j}{s_{\hat{b}_j}} \leq z_{1-\alpha/2} \right) \\ &= P \left(\hat{b}_j + z_{\alpha/2} s_{\hat{b}_j} \leq b_j \leq \hat{b}_j + z_{1-\alpha/2} s_{\hat{b}_j} \right). \end{aligned}$$

Therefore, an approximate confidence interval at $1 - \alpha$ is

$$\left[\hat{b}_j - z_{1-\alpha/2} s_{\hat{b}_j}, \hat{b}_j + z_{1-\alpha/2} s_{\hat{b}_j} \right].$$

The most common CI is at 95% ($\alpha = 5\%$), with $z_{0.975} = 1.96$, thus

$$\left[\hat{b}_j - 1.96 \times s_{\hat{b}_j}, \hat{b}_j + 1.96 \times s_{\hat{b}_j} \right].$$

Logistic regression: confidence intervals

Example: IC(95%) for b_1 (Pages.Week)

$$[0.089 \pm 1.96 \times 0.036] = [0.019, 0.159]$$

For all the others,

```
> confint.default(mod1)
```

	2.5 %	97.5 %
(Intercept)	-4.79838189	0.08504130
Pages.Week	0.01942094	0.15913334
AvPages.Week	-0.02717169	0.07417085
Age25-29	-1.67637430	3.05199536
Age30-39	-1.53715743	3.46362841
Age40-49	-5.68224201	-0.28323727
Age50-59	-1.20279550	3.37386306
Age60+	-2.16920353	2.78704580

Logistic regression: profile confidence intervals

Unlike for the regression, exact confidence intervals cannot be built for logistic regression. Slightly more precise confidence intervals than the normal one can be built using profile likelihood (outside the scope of this course). These intervals are asymmetric. When available, they should be preferred.

Example: By default R gives the profile confidence intervals

```
> confint(mod1)
Waiting for profiling to be done...
              2.5 %      97.5 %
(Intercept) -5.14449579 -0.15334930
Pages.Week   0.02606627  0.16734103
AvPages.Week -0.02582757  0.07933927
Age25-29     -1.71426098  3.16597735
Age30-39     -1.47409600  3.69456739
Age40-49     -6.31213265 -0.58343865
Age50-59     -1.14607725  3.56240074
Age60+       -2.20859149  2.88615495
```

Logistic regression: model diagnostics

Like for the regression, model assumptions have to be checked using residuals. The assumptions are

- ▶ The outcomes are distributed according to a Bernoulli law

$$P(Y_i = 1|x) = p(x), \quad P(Y_i = 0|x) = 1 - p(x).$$

- ▶ The observations are independent.

How can we define residuals for 0/1 data?

Logistic regression: model diagnostics

Two types of residuals are used

- ▶ Deviance residuals

$$d_i = s_i \sqrt{-2 \{y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)\}}$$

where $s_i = 2y_i - 1$ (i.e. $s_i = 1$ if $y_i = 1$, $s_i = -1$ if $y_i = 0$).

- ▶ Pearson residuals

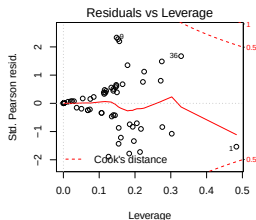
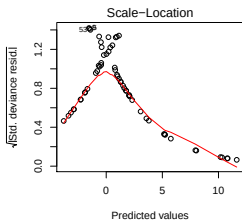
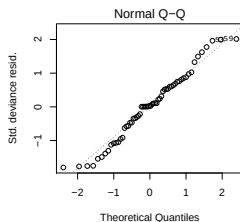
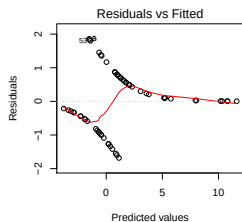
$$r_i = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}.$$

Deviance residuals have slightly better theoretical properties than Pearson residuals, though they can both be used with the same aim (depending on the software you use).

Like for regression, if the model assumptions are validated by the data, the residuals should be distributed (approximately) as a $N(0, 1)$.

Logistic regression: model diagnostics

The 4 diagnostic plots are the same as for the regression.



Logistic regression: model diagnostics

- ▶ Top-left: deviance residuals (d_i) versus predicted values ($\hat{\eta}_i$).
- ▶ Top-right: QQ-plot normal versus d_i .
- ▶ Bottom-left: $\sqrt{|d_i|}$ versus $\hat{\eta}_i$.
- ▶ Bottom-right: r_i versus a leverage measure.

Logistic regression: model diagnostics

Top-Left: the upper points are $y_i = 1$. They are predicted as 1 when $\hat{\eta}_i$ is large (right side of the plot). Close to $x = 0$, the residuals increases, it is an uncertain zone. Same for lower points (symmetric). This is a classical pattern for logistic regression.

Top-right: QQ-plot should present the same pattern as in the regression case.

Bottom-left: results from the absolute value. The variance is higher when $\hat{\eta}_i$ is close to 0 (uncertain zone). It is very similar to the top-left plot.

Bottom-right: use the same as for the regression. It helps to detect influential points.

Statistical modeling

Module 11

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Autumn 2018

Generalized Linear Models

Regression and logistic regression belongs to a larger family of models called Generalized Linear Models (GLM). They can be used the same way and the choice mainly depends on the type of the response y .

- ▶ Regression: y is numerical, normally distributed (symmetric and continuous or when discontinuity is not important).
- ▶ Logistic: y is binary (0/1).
- ▶ Poisson: y is a count (0, 1, 2, etc.).
- ▶ Gamma: y is positive (asymmetric distribution with on $[0, \infty[$).

Other models have been developed depending on the specific application.

GLM: count data

Example: Medical care demand study: y is the number of visits to a physician office ofp as a function of

- ▶ Number of hospital stays: `hosp`, x_1 , num.
- ▶ Health: `health`, x_2 , (average/excellent/poor).
- ▶ Number of chronic conditions: `numchron`, x_3 , num.
- ▶ gender: x_4 (female/male)
- ▶ Number of years of education: `school`, x_5 , num.
- ▶ private insurance: `privins`, x_6 (no/yes)

Source: <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf>

GLM: count data

For a set of variables x , the linear predictor is (like for the logistic regression),

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p$$

Given x , the response $y(x)$ is assumed to be distributed according to a Poisson distribution with mean $\lambda(x)$,

$$y(x) \sim \text{Pois}(\lambda(x)).$$

In details,

$$P(y(x) = k) = \frac{\lambda(x)^k}{k!} e^{-\lambda(x)}, \quad k \geq 0.$$

The mean $\lambda(x)$ and the linear predictor $\eta(x)$ are linked by the log-link function

$$\eta(x) = \log \lambda(x)$$

GLM: count data

With R, the model is fitted as

```
> mod <- glm(ofp ~ hosp + health + numchron + gender + school + privins, family=poisson, data=dt)
> summary(mod)
[...]
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.028874	0.023785	43.258	<2e-16 ***
hosp	0.164797	0.005997	27.478	<2e-16 ***
healthexcellent	-0.361993	0.030304	-11.945	<2e-16 ***
healthpoor	0.248307	0.017845	13.915	<2e-16 ***
numchron	0.146639	0.004580	32.020	<2e-16 ***
gendermale	-0.112320	0.012945	-8.677	<2e-16 ***
school	0.026143	0.001843	14.182	<2e-16 ***
privinsyes	0.201687	0.016860	11.963	<2e-16 ***

```
---
[...]
```

GLM: count data

The estimated linear predictor is

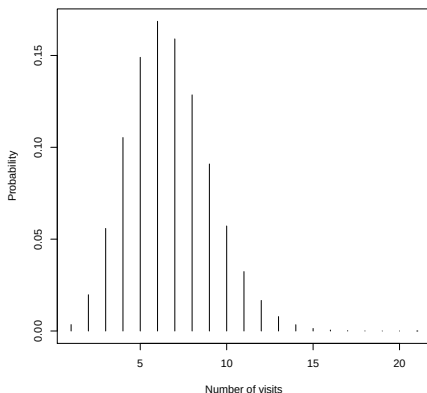
$$\hat{\eta}(x) = 1.03 + 0.16x_1 - 0.36\delta_{\text{excl}}(x_3) + 0.25\delta_{\text{poor}}(x_3) \\ + 0.15x_4 - 0.11\delta_{\text{male}}(x_5) + 0.03x_6 + 0.20\delta_{\text{yes}}(x_7)$$

```
> dt[1,c(6,7,8,15,17,20)] ## x
  hosp health numchron gender school privins
1    1 average         2  male      6     yes
> predict(mod, type="link")[1] ## eta(x)
1
1.733175
> predict(mod, type="response")[1] ## y(x)
1
5.658592
```

The prediction can be rounded to 6 in practice.

GLM: count data

In more details, the model tells us that the number of visits of p for the population of subjects with characteristics x (previous slide) is distributed as a Poisson distribution with mean 5.66, that is



GLM: gamma distribution

Example: Insurance study (Sweden 1977). The response y is the total value payment `Payment`. It should be explained as function of

- ▶ The car model `Make`: x_1 , categorical with 9 levels,
- ▶ The number of insured in policy-years `Insured` (in log-scale): x_2
- ▶ The bonus claim `Bonus`: x_3 , (Y/N).

Source: <http://people.stat.sc.edu/Hitchcock/stat599chap7Rcode.txt>

GLM: gamma distribution

For a set of variables x , the linear predictor is (like always)

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p$$

Then, given x , the response $y(x)$ is assumed to be distributed as a Gamma distribution with scale $\theta(x)$ and shape r

$$y(x) \sim Ga(\theta(x), r).$$

In details, the density function of y given x is

$$f(y; \theta(x), r) = \frac{1}{\Gamma(r)\theta(x)^r} y^{r-1} e^{-y/\theta(x)}.$$

GLM: gamma distribution

The mean and variance of y given x are

$$E(y) = r\theta(x), \quad \text{Var}(y) = r\theta(x)^2.$$

The link between $\eta(x)$ and $\theta(x)$ is the inverse:

$$\eta(x) = 1/\theta(x).$$

The shape parameter r is estimated as a constant (independent of x).

GLM: gamma distribution

The fitted model is

```
> mod <- glm(Payment ~ log(Insured) + Make + Bonus, family = Gamma, motor)
> summary(mod)
```

[...]

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.191e-06	8.302e-07	7.456	1.08e-12	***
log(Insured)	-5.532e-07	1.117e-07	-4.951	1.27e-06	***
Make2	7.127e-06	2.114e-06	3.371	0.000852	***
Make3	9.335e-06	2.534e-06	3.684	0.000275	***
Make4	8.831e-06	2.691e-06	3.282	0.001159	**
Make5	1.040e-05	2.775e-06	3.747	0.000217	***
Make6	7.292e-06	2.102e-06	3.470	0.000602	***
Make7	1.836e-05	4.317e-06	4.253	2.87e-05	***
Make8	2.583e-05	6.086e-06	4.244	2.97e-05	***
Make9	-1.148e-06	7.221e-07	-1.590	0.112887	
Bonus	1.378e-07	3.531e-08	3.904	0.000118	***

(Dispersion parameter for Gamma family taken to be 1.126119)

GLM: gamma distribution

The prediction is the mean, that is

$$\hat{y}(x) = \hat{\mu}/\hat{\eta}(x).$$

In the example, the prediction is

```
> motori[1,c(3:5,7)] ## x
  Bonus Make Insured Payment
1      1      1 455.13 392491
> 1.126/predict(mod, type="link")[1] ## y.hat
      1
382634.1
```

because of the estimate of the shape parameter $\hat{\mu} = 1.126$.

Note: in practice, transforming the response y to its log and using a regression is often preferred to a gamma model because it is easier to interpret at the end.

GLM

Generalized Linear Models encompass a lot of models and can be used in a coherent manner, similar to linear regression, i for many types of outcomes:

- ▶ Numerical and real: linear regression (normal GLM)
- ▶ Counts: Poisson GLM
- ▶ Positive: log-linear regression or gamma GLM
- ▶ Binary: logistic regression (binomial GLM)
- ▶ Categorical: multinomial model
- ▶ $[0, 1]$ data (like %): beta regression.

and others.

Statistical modeling

Module 12: Revisions

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Directives for the exam

They are on the chamilo page.

- ▶ One final written exam at the end of the session (2h45).
- ▶ Restricted documentation: you are allowed to bring 12 A4 pages (6 sheets) prepared on your own. No writing style limitation.
- ▶ Exam type: open questions (maybe one or two multiple choices for fun).
- ▶ Includes: interpretations of results (e.g. of R outputs), analyses of situations (e.g. making recommendations in some situations), some calculations that can be performed with a simple calculator. If the exam is not testing your memory and your R programming skills, it is however clear that you are expected to be familiar with the functions treated in exercises and that you know the concepts treated in class. Otherwise, interpretations, analyses, and recommendations are useless. This is why the "limited authorized documentation" is large enough.

What we have studied 1

General and introduction:

- ▶ What is a model
- ▶ Dependence between 2 variables:
- ▶ Scatter plots
- ▶ Correlations: Pearson, Spearman, Kendall's
- ▶ Covariance

The regression equation and its assumptions

- ▶ Some theory: OLS and sum-of-squares, matrix notation, sum of squares decomposition
- ▶ Prediction equation
- ▶ Quality: R^2 , adjusted R^2 , plot observation versus fitted values

What we have studied 2

Categorical variables

- ▶ Dummy variables coding
- ▶ Interpretation of the coefficients, contrasts
- ▶ Interactions: cat-cat, num-num, cat-num, higher order interactions

Uncertainty:

- ▶ Standard error (standard deviation of the error)
- ▶ Prediction intervals
- ▶ Notions of model building

What we have studied 3

Inference:

- ▶ the fundamental concept, population, sample.
- ▶ F-Tests:
 - ▶ Model significance
 - ▶ Anova table
 - ▶ Model comparisons (and its link to the two previous ones)
 - ▶ Anova of type I, II and III
- ▶ Inference on the coefficients:
 - ▶ standard errors of the estimate, t-value and t-test
 - ▶ confidence intervals
 - ▶ test on linear combinations of coefficients

What we have studied 4

Regression assumptions:

- ▶ $e_i \sim N(0, \sigma^2)$ + independence
- ▶ Standardized residuals
- ▶ Diagnostics plots:
 - ▶ Residuals vs fitted values
 - ▶ QQ plots
 - ▶ Scale-location
 - ▶ Residuals versus leverage
- ▶ Notion of outliers and influential values.

Special topics:

- ▶ Moderation test
- ▶ Mediation test(s), total, direct and indirect effects.
- ▶ Association versus causality
- ▶ Analysis of collinearity, VIF.
- ▶ Variable selection: test based and AIC based approaches.

What we have studied 5

Logistic regression

- ▶ Binary outcome: graphical analysis
- ▶ Linear predictor, logit link function, predicted probability, predicted outcome.
- ▶ Interpretation of the coefficients (values on the linear scale, sign on the probability scale)
- ▶ Interactions
- ▶ Interpretation of predicted probability (equiv. prediction intervals)
- ▶ Estimation: Maximum likelihood and log-likelihood (theory)

Quality

- ▶ Confusion matrix,
- ▶ Accuracy, Sensitivity, specificity,
- ▶ ROC curve, AUC

What we have studied 6

Inference

- ▶ Comparison of models: LRT (incl. degree of freedom)
- ▶ Normal test on the coefficients
- ▶ Confidence intervals: normal based, others (profile likelihood)

Model assumptions

- ▶ Bernoulli distribution
- ▶ Deviance and Pearson residuals
- ▶ 4 diagnostics plots

GLM: concept and examples (type of outcomes, different distributions)

About the exam...

Possible calculations by hand (simple scientific calculator)

- ▶ Compute a prediction of a model given the coefficients (e.g. R output of a logistic regression), giving all the intermediate steps
- ▶ Compute a prediction interval
- ▶ Compute a confidence interval
- ▶ Identify an outlier given the number of data
- ▶ Compute an accuracy, sensitivity, specificity from a confusion matrix
- ▶ In any case: additions, subtractions, multiplications, divisions, power, exponential, logarithm, etc.

Other productions:

- ▶ Graphs of mediation and moderation in a given situation.
- ▶ Being able to complete a ROC curve given some results (e.g. specificities and sensitivities for several thresholds)
- ▶ Coding in dummy variables and interactions.
- ▶ Any other work on graphs showing you know what you are talking about.

About the exam...

Analysis:

- ▶ Be able to interpret any R output of the types that was used in the course, in the exercises, or that is similar to these.
- ▶ From some descriptive analysis (e.g. scatter plot) be able to anticipate some of the results of a further model, keeping the limits of these approach in mind.
- ▶ From R output, be able to make several steps of an analysis (e.g. a mediation analysis, a variable selection, etc.).

Discussions:

- ▶ Given a situation, data, etc. be able to propose and justify a type of analysis in full details (regression, logistic regression, which variables, why, moderation, mediation, etc.)
- ▶ Make proposition of improvements and what can be expected from it (e.g. change of variable, etc.)
- ▶ Others...

About the exam...

Type of exam questions

- ▶ Open questions
- ▶ Direct and without trap (or very very obvious ones, or not on purpose...)
- ▶ If you are not sure, write what you think is correct and explain.
- ▶ Technical questions (e.g. a calculation) are here to allow you to show that you know the concept and are able to do it. A small error in the numbers will not be severely sanctioned (but still do your best otherwise, little by little you lose points).